

# CCNA Exam C

## Reorganized to Follow Jeremy's IT Lab CCNA 200-301 Course

74 items - questions, correct answers, and source explanations

**What changed:** The original Exam C items are reorganized into Jeremy's course order so you can review Boson questions immediately after the matching Jeremy topic. The wording of the source explanations is preserved as closely as practical; obvious PDF/OCR text-layer artifacts are normalized for readability.

**Lab items:** Original questions 1-4 are NetSim lab tasks. The source provides the task prompt but no separate correct-answer/explanation section, so they are reproduced as lab tasks only.

**Cross-topic items:** A few integrated questions touch multiple subjects. Each appears once under the closest primary Jeremy topic to avoid duplicates.

**Selectable/searchable:** All rewritten question, answer, and explanation text in this document is real text, not page screenshots.

## Course-Order Index

Only Jeremy sections that contain at least one Exam C item are listed. Original Exam C question numbers are retained for traceability.

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## Day 2 - Interfaces & Cables

Included Exam C items: Q31

### Original Exam C Question 31

Which of the following is an IEEE Ethernet standard that uses single-mode fiber cable? (Select the best answer.)

- A. 1000BaseSX
- C. 1000BaseCx
- D. 1000BaseZxX

**Correct Answer: B**

Explanation

The 1000BaseLX standard primarily uses single-mode fiber cable. The 1000BaseLX gigabit Ethernet standard is a subset of the Institute of Electrical and Electronics Engineers (IEEE) 802.3z standard. The 1000BaseLX standard is typically used to provide gigabit Ethernet over distances of up to 3.1 miles (5 kilometers). Fiber-optic cables are made up of a glass core, reflective cladding, and a buffer to provide rigidity. The outside of the cable is covered with a protective jacket, which is typically made of rubber. Data is transmitted through fiber-optic cable as pulses of light. Because fiber-optic cables transmit data as pulses of light, they are not susceptible to electromagnetic interference (EMI). EMI can result when electrical current from an external source, such as lights or appliances, interferes with the transmission of data via electrical pulses. ST, SC, and LC connectors are used to terminate fiber connections. The 1000BaseCx IEEE standard does not primarily use single-mode fiber cable. Instead, twinaxial cable is used for 1000BaseCx installations. Twinaxial cables are made up of dual copper wires that are individually wrapped by an insulator and a metallic shield. Both cables are sheathed in a plastic outer covering that protects the cable. The 1000BaseCX standard, which is a subset of the IEEE 802.3z standard, is used for installations covering relatively short distances of up to 82 feet (25 meters). The 1000BaseSX IEEE standard does not primarily use single-mode fiber cable. Instead, 1000BaseSX primarily uses multimode fiber. Multimode fiber transmits signals with the use of light-emitting diodes (LEDs) on an 850-nanometer (nm) wavelength. The 1000BaseSX standard, which is a subset of the IEEE 802.3z standard, is used for installations covering distances of up to 1,640 feet (500 meters). The 1000BaseZX standard is a proprietary specification that uses single-mode fiber cable. The 1000BaseZX standard is typically used to provide gigabit Ethernet across distances up to 43.5 miles (70 kilometers).

## Days 5-6 - Ethernet LAN Switching

Included Exam C items: Q20

### Original Exam C Question 20

Which of the following best describes the Ethernet FCS field? (Select the best answer.)

- It contains a 2-byte representation that includes the number of bytes that precede and follow it.
- B. It contains the protocol that receives the frame after Ethernet processing.
- C. It contains a payload header and the frame payload.
- D. It contains a 4-byte CRC value.

**Correct Answer: D**

Explanation

The Ethernet frame check sequence (FCS) field contains a 4-byte cyclic redundancy check (CRC) value. Ethernet, which operates at the Data Link layer of the Open Systems Interconnection (OSI) networking model, is calculated by the frame's sender. When the frame arrives at its destination, the receiver recalculates the CRC and compares it to the value that is stored in the FCS field. If the values match, the frame is sent to the next-layer protocol. If the values do not match, the frame is typically discarded and the interface FCS-err value is incremented. A large number of FCS errors could indicate a faulty interface or damaged cable. Methods similar to CRCs are used at several other layers of the OSI networking model. The CRCs used by Ethernet and the Institute of Electrical and Electronics Engineers (IEEE) 802.3 standard are a form of checksum. Network interface cards (NICs), which operate at the Physical layer of the OSI networking model, can also use checksums as a

form of digital error detection. Internet Protocol (IP), which operates at the Network layer of the OSI networking model, uses a header checksum to protect the header. Checksums used in protocols that operate at layers higher than the Network layer typically protect both the transmission's header and its data. The Ethernet type field, not the FCS field, contains the protocol that receives the frame after Ethernet processing. This field is 2 bytes in length. The type field is not present in the IEEE 802.3 standard on which Ethernet is based. Instead, the 802.3 standard contains a length field at the location of the type field. This length field is a 2-byte representation of the number of bytes that follow the length field as well as the number of bytes that precede this field. The Ethernet data field, not the FCS field, contains the frame payload. In the 802.3 standard, this field also contains an 802.2 payload header.

## Days 7-8 - IPv4 Addressing

Included Exam C items: Q14

### Original Exam C Question 14

Which of the following is a troubleshooting command that can be used to display basic information about the computer's network interfaces and to perform various utility functions on those interfaces? (Select the best answer.)

- A. netstat
- B. ipconfig
- C. ping
- D. tracert

**Correct Answer: B**

#### Explanation

The ipconfig command is an administrative troubleshooting tool that can be used to display basic information about the computer's network interfaces and to perform various utility functions on those interfaces. For example, typing ipconfig at a Windows command-line interface (CLI) command prompt will display information such as the Internet Protocol (IP) address, subnet mask, and default gateway for each network adapter bound to Transmission Control Protocol/IP (TCP/IP) for the computer. The following is an example of what might be displayed when you issue the ipconfig command: C:\Users\Owner>ipconfig /all Windows IP Configuration Host Name... : boson.com Primary Dns Suffix : Node Type : Hybrid IP Routing Enabled. : No WINS Proxy Enabled. : NO DNS Suffix Search List. : boson.com Ethernet adapter Ethernet: Connection-specific DNS Suffix : boson.com : Intel(R) Ethernet Connection (2) 1219-V Physical Address. : 00-23-24-B5-45-7D Enabled. DHCP : Yes Autoconfiguration Enabled : Yes Link-local IPv6 Address : fe80::572e:839d:fc5a:7ca9%10(Prefered) Ipv4 Address. : 192.168.86.110(Prefered) Subnet Mask... : 255.255.255.0 Lease Obtained. : Saturday, April 1, 2023 4:20:08 AM Lease Expires : Monday, April 3, 2023 2:15:55 PM Default : 192.168.86.1 DHCP Server... : 192.168.86.98 DHCPv6 IAID... : 167781156 DHCPv6 Client DUID. : 00-01-00-01-26-93-EB-2E-00-23-24-B5-45-7D DNS Servers... : 192.168.86.98 NetBIOS over Tcpi. : Enabled From the information displayed, you can see that the computer has the IP address 192.168.86.110, which was obtained from the Dynamic Host Configuration Protocol (DHCP) server at 192.168.86.98. In addition, the computer has been configured to use the default gateway address 192.168.86.1 and the Domain Name System (DNS) server address 192.168.86.98. The ipconfig command is helpful for troubleshooting a network interface card (NIC) because it enables you to verify that the IP address is configured properly. Verifying that a workstation has an IP address by issuing the ipconfig /all command should be one of the first steps in troubleshooting that workstation's network connection. In addition, you can use the Ipconfig tool to modify DNS settings. For example, you can use the /flushdns switch to erase the contents of the local DNS cache and the /registerdns switch to communicate with a DNS server to update DNS settings. Finally, you can use the ipconfig /release command followed by the ipconfig /renew command to force a workstation to delete an IP address configuration and retrieve a new IP address lease from a DHCP server. You should not use the tracert command to determine the network properties of all NICs installed in a computer. You can use tracert to determine the route taken by packets across an IP network as well as to troubleshoot router or bridge functionality. The tracert command sends Internet Control Message Protocol (ICMP) echo packets to a destination on the network and then examines the Time Exceeded messages returned by intermediate routers as well as the echo reply message returned by the destination. By using the tracert command to determine where the packet has stopped on the network, you can isolate any potentially faulty routers or connections. You should not use the netstat command to determine the network properties of all NICs installed in a computer. You can use netstat to view network interface statistics, such as what Transmission Control Protocol (TCP) connections view are active, and to a list of

listening ports. You should not use the ping command to determine the network properties of all NICs installed in a computer. You would use the ping command to determine whether a particular host is reachable over an IP network or to test the functionality of the computer's NIC. The ping command approximates the round-trip time, as well as any packet loss, and displays a summary of the results.

## Day 9 - Switch Interfaces

Included Exam C items: Q17

### Original Exam C Question 17

Which of the following do not indicate a duplex mismatch on an Ethernet LAN? (Select the best answer.)

- A. late collisions
  - B. FCS errors
  - C. alignment errors
  - D. baby giants
- Correct Answer(s): D Explanation:

Of the available choices, baby giants do not indicate a duplex mismatch on an Ethernet local area network (LAN). A baby giant is an Ethernet frame that is up to 1,600 bytes in length. The default maximum transmission unit (MTU) size for Ethernet frames is 1,500 bytes, not including the Ethernet header and the cyclic redundancy check (CRC) trailer, which add 18 bytes to the frame. Baby giant frames are slightly larger than an Ethernet frame. These can occur if you use Q-in-Q encapsulation, Multiprotocol Label Switching (MPLS), or any other feature that adds to the size of an Ethernet frame.

A late collision is an Ethernet collision that occurs after 512 bits of a frame have already been transmitted. Typically, collisions are detected within a 51.2-microsecond time frame, or 512 bits. Thus an Ethernet cable that is too long might create late collisions. In addition, a half-duplex port that is connected late collisions the side of the connection.

a full-duplex port can report on half-duplex Runts, Frame Check Sequence (FCS) errors, and alignment errors can all indicate a duplex mismatch on an Ethernet LAN. Although the half-duplex side of a duplex mismatch will report late collisions, the full-duplex side will report different errors, such as runts, FCS errors, and alignment errors. A runt is a frame that is fewer than 64 bytes and has a bad FCS.

Reference:

Boson CCNA Curriculum 200-301, Module 1: Network Fundamentals, Troubleshooting Interfaces and Cabling CCNA 200-301 Official Cert Guide, 2nd Edition, Volume 1, Chapter 7: Configuring and Verifying Switch Interfaces, Common Layer 1 Problems on Working Interfaces Cisco Press: Ethernet Switching: Ethernet Frame Fields (7.1.4)0 Cisco: Troubleshooting LAN Switching Environments: Configuration Issues0 Category: Network Fundamentals

**Correct Answer: D**

Explanation

Of the available choices, baby giants do not indicate a duplex mismatch on an Ethernet local area network (LAN). A baby giant is an Ethernet frame that is up to 1,600 bytes in length. The default maximum transmission unit (MTU) size for Ethernet frames is 1,500 bytes, not including the Ethernet header and the cyclic redundancy check (CRC) trailer, which add 18 bytes to the frame. Baby giant frames are slightly larger than an Ethernet frame. These can occur if you use Q-in-Q encapsulation, Multiprotocol Label Switching (MPLS), or any other feature that adds to the size of an Ethernet frame. A late collision is an Ethernet collision that occurs after 512 bits of a frame have already been transmitted. Typically, collisions are detected within a 51.2-microsecond time frame, or 512 bits. Thus an Ethernet cable that is too long might create late collisions. In addition, a half-duplex port that is connected late collisions the side of the connection. to a full-duplex port can report on half-duplex Runts, Frame Check Sequence (FCS) errors, and alignment errors can all indicate a duplex mismatch on an Ethernet LAN. Although the half-duplex side of a duplex mismatch will report late collisions, the full-duplex side will report different errors, such as runts, FCS errors, and alignment errors. A runt is a frame that is fewer than 64 bytes and has a bad FCS.

# Days 11-12 - Routing Fundamentals, Static Routing & Life of a Packet

Included Exam C items: Q8, Q21, Q23

## Original Exam C Question 8

You issue the show ip route command on RouterA and receive the following partial output:

RouterA#show ip route

<output omitted> Gateway of last resort is 10.19.54.28 to network 10.140.0.0 O E2 172.150.0.0 via 10.19.54.6, 0:01:00, FastEthernet1/2 What does the 110 in the output indicate? (Select the best answer.)

- A. the routing protocol
- B.
- C. the hop count
- D. the metric

**Correct Answer: B**

Explanation

The 110 in the output indicates the administrative distance (AD); the AD is the first number inside the brackets. When multiple routes to a network exist, a router prefers the routing protocol with the lowest AD. The following list contains the most commonly used default ADs: Route Source AD Directly connected route 0 | Internal EIGRP IGRP 100 OSPF IS-IS RIP External EIGRP connected which has of 0, is other the A directly route, an AD preferred over any route to same network. If a link to a directly connected network goes down, the route with the next lowest AD will be used. Static routes, which have an AD of 1 by default, are preferred after directly connected routes. The 0 in the output indicates the routing protocol, which, in this scenario, is Open Shortest Path First (OSPF). An R in the output would indicate a Routing Information Protocol (RIP) route, and aD in the output would indicate an Enhanced Interior Gateway Routing Protocol (EIGRP) route. Although an AD of 110 would seem to indicate that the routing protocol is OSPF, the AD of a route can be modified from the default value. For example, you could create a static route with an AD of 110. The 5 in the output indicates the metric; the metric is the second number inside the brackets. OSPF uses cost as a metric and calculates cost based on the bandwidth of an interface: the higher the bandwidth, the lower the cost. When two OSPF paths exist to the same destination, the router will choose the OSPF path with the lowest cost. The 110 in the output does not indicate hop count. OSPF uses cost, not hop count, as a metric. By contrast, RIP uses hop count as a metric.

## Original Exam C Question 21

Ol jay Router2\_ net Router1 You administer the network in the exhibit.

You issue the show ip route command on Router1 and receive the following partial output:

Router1#show ip route

<output omitted> Gateway of last resort is not set 10.0.0.0/30 is subnetted, 1 subnets via Ethernet0/0/0 10.10.10.0

192.168.1.2, 00:00:00, 192.168.1.0/30 is subnetted, 1 subnets Cc 192.168.1.0 is directly connected, Ethernet0/0 You issue the ip route 10.10.10.0 255.255.255.252 ethernet 0/0 command on Router1.

Which of the following will happen next? (Select the best answer.)

- A. The static route will be used as a backup if the OSPF route goes down.
- B. Routing loops or performance problems will occur because points back to itself.
- C. The OSPF route will be replaced with a static route.
- D. The gateway of last resort will be the Ethernet 0/0 interface on

**Correct Answer: C**

Explanation

The Open Shortest Path First (OSPF) route will be replaced with a static route to the 10.10.10.0/30 network if you issue the ip route 10.10.10.0 255.255.255.252 ethernet 0/0 command on Router1. The output of the show ip route command in this scenario indicates that Router1 has an OSPF path to the 10.10.10.0/30 network. OSPF is a link-state routing protocol, which means that it learns routes dynamically. If a router has a static route and an OSPF route to the same destination network, the static route will be preferred over the dynamic route because static routes have a lower administrative distance (AD) than routing protocols. Static routes are manually configured by an administrator. They are typically more secure than dynamically learned routes because only an administrator can introduce changes to the route. In addition, statically configuring routes on a router in a stub network can help reduce the traffic load that must be processed by the router. A stub network is a network that

does not require direct knowledge of other networks that lie beyond a default gateway. Dynamic routing protocols such as OSPF increase demand on router resources because they use formulas, such as the Dijkstra Shortest Path First (SPF) algorithm, to calculate the best path to a destination. To configure a static route on a Cisco router, you should issue the `ip route prefix mask {ip-address | interface}` command in global configuration mode. The prefix value is the network to which an incoming data packet is destined. The mask value is the subnet mask for the destination network. The ip-address value is the Internet Protocol (IP) address of the next-hop router, and the interface value is the local interface to which the packets should be sent. Neither routing loops nor performance problems will occur, because Router1 is not pointing back to itself and there is no evidence of a discontinuous subnet configuration in the topology. Routing loops occur when a topology change or a delayed routing update results in two routers pointing to each other as the next hop to a destination. In this scenario, the `ip route 10.10.10.0 255.255.255.252 ethernet 0/0` command configures Router1 to send any traffic that is destined for the 10.10.10.0/30 network out Router1's Ethernet 0/0 interface, not to Router1. You can also configure the destination for a static route with a next-hop IP address instead of an outgoing interface. If Router1's Ethernet 0/0 interface is configured with an IP address of 192.168.1.1, an `ip route` command that uses the next-hop IP address as a destination would be `ip route 10.10.10.0 255.255.255.252 192.168.1.2`. This command configures to send any traffic that is destined for the 10.10.10.0/30 network to the directly connected interface on Router2, which is configured with the IP address of 192.168.1.2. Although the performance of link-state routing protocols such as OSPF is optimized when a topology is deployed with a hierarchical IP addressing scheme, there is no indication in this scenario that the topology is configured in a discontinuous way. If the 192.168.1.4/30 network were attached to the Ethernet 0/1 interface of Router3, the classful 192.168.1.0/24 network would be discontinuous in the topology because it would be separated by the 10.10.10.0/30 network that lies between Router2 and Router3. The static route will not be used as a backup if the OSPF route goes down, because the static route will replace the OSPF route in the routing table. If the static route were to be removed from the configuration, the OSPF route would again be inserted into the routing table. By default, OSPF has an AD of 110. A static route, on the other hand, has a default AD of 1. Because the static route has a better AD, the static route will replace the OSPF route in the routing table on Router1. Static routes remain in the routing table even if the path to the destination goes down. If the static route were configured so that it had an AD higher than 110, it could be used as a backup if the OSPF route went down. For example, you could issue the command `ip route 10.10.10.0 255.255.255.252 ethernet 0/0 120` to configure a floating static route with an AD of 120. The gateway of last resort will not be the Ethernet 0/0 interface on Router1. The `ip route 10.10.10.0 255.255.255.252 ethernet 0/0` command configures to send any traffic that is destined for the specific 10.10.10.0/30 network out the Ethernet 0/0 interface on Router1. A gateway of last resort, which is also known as a default gateway or a default route, is an interface or next-hop IP address to which the router will send any traffic that is destined for a path that is not listed in the routing table. To create a default route, you should issue the `ip route 0.0.0.0 0.0.0.0 destination` command, where destination is the next-hop IP address or interface to which the router should forward the data packet.

## Original Exam C Question 23

In which of the following situations does a router use AD values to determine route selection? (Select the best answer.)  
when multiple routes to the same destination network are received, and each of these routes is received from a different routing protocol  
when multiple routes to different destination networks are received, and each of these routes is received from a different routing protocol  
when multiple routes to different destination networks are received, and all of these routes are received from the same routing protocol  
when multiple routes to the same destination network are received, and all of these routes are received from the same routing protocol

**Correct Answer: A**

### Explanation

A router uses administrative distance (AD) values to determine route selection when multiple routes to the same destination network are received, and each of these routes is received from a different routing protocol. Lower ADs are preferred over higher ADs. The following list contains the most commonly used ADs: Route Source Directly connected route Static route EIGRP summary route eBGP Internal EIGRP IGRP OSPF IS-IS RIP External EIGRP iBGP Unknown A router uses metrics to determine route selection when multiple routes to the same destination network are received, and all of these routes are received from the same routing protocol. Each routing protocol uses different metrics. For example, Routing Information Protocol (RIP) uses hop count as a metric, Open Shortest Path First (OSPF) uses cost as a metric, and Enhanced Interior Gateway Routing Protocol (EIGRP) uses bandwidth and delay by default as a composite metric. When a routing protocol contains multiple routes to the same destination network, a router prefers the route with the lowest metric. A router uses prefix lengths to determine route selection when multiple routes to different destination networks are received, regardless of the routing protocol. When multiple routes to overlapping networks exist, a router will prefer the most specific route, which is the route with the longest prefix match. For example, if a router has a packet destined to 10.1.1.1, it will prefer a route to 10.1.1.0/24 over a route to 10.1.1.0/16 and it will prefer a route to 10.1.1.0/30 over a route to 10.1.1.0/24.

## Days 13-15 - Subnetting

Included Exam C items: Q29, Q43, Q67

### Original Exam C Question 29

You are creating a network addressing scheme for a new branch office. The branch office will require 4,000 addresses. Your supervisor has instructed you to conserve IPv4 address space.

Which of the following subnet masks should you use for the branch office? (Select the best answer.)

- A. 255.255.248.0
- B. 255.255.0.0
- C. c. 255.255.128.0
- D. 255.255.240.0
- E. 255.255.192.0

**Correct Answer: D**

#### Explanation

Of the available subnet masks, you should use the 255.255.240.0 subnet mask for the branch office. To determine how many host addresses can be allocated within a subnet, use the formula  $2^n - 2$ , where  $n$  is the number of bits in the host portion of the 32-bit Internet Protocol version 4 (IPv4) address. A subnet mask of 255.255.240.0 converts to a binary equivalent of 11111111.11111111.11110000.00000000. The twenty 1s in the binary subnet mask are network bits. The twelve 0s in the binary subnet mask are host bits. Therefore, the formula to calculate available hosts in a 255.255.240.0 subnet becomes  $2^{12} - 2$ , which equals 4,094; this is sufficient for the new branch office. Subnet masks can be expressed in dotted decimal notation, such as 255.255.240.0, or in Classless Inter-Domain Routing (CIDR) notation. CIDR notation simplifies subnet masks by indicating the number of network bits in a subnet. For example, the subnet mask 255.255.240.0 has 20 network bits and 12 host bits; therefore, the CIDR notation for the 255.255.240.0 subnet mask is /20. Although it is important to learn how to calculate the relationship between subnet masks, CIDR notation, and valid host addresses, the following list demonstrates the relationship between them: © 255.255.0.0 (/16) subnet mask contains  $2^{16} - 2 = 65,534$  host addresses © 255.255.128.0 (/17) subnet mask contains  $2^{15} - 2 = 32,766$  host addresses © 255.255.192.0 (/18) subnet mask contains  $2^{14} - 2 = 16,382$  host addresses (/19) subnet mask contains  $2^{13} - 2 = 8,190$  host addresses (/20) subnet mask contains  $2^{12} - 2 = 4,094$  host addresses © 255.255.248.0 (/21) subnet mask contains  $2^{11} - 2 = 2,046$  host addresses © 255.255.252.0 (/22) subnet mask contains  $2^{10} - 2 = 1,022$  host addresses (/23) subnet mask contains  $2^9 - 2 = 510$  host addresses © 255.255.255.0 (/24) subnet mask contains  $2^8 - 2 = 254$  host addresses © 255.255.255.128 (/25) subnet mask contains  $2^7 - 2 = 126$  host addresses © 255.255.255.192 (/26) subnet mask contains  $2^6 - 2 = 62$  host addresses © 255.255.255.224 (/27) subnet mask contains  $2^5 - 2 = 30$  host addresses © 255.255.255.240 (/28) subnet mask contains  $2^4 - 2 = 14$  host addresses © 255.255.255.248 (/29) subnet mask contains  $2^3 - 2 = 6$  host addresses © 255.255.255.252 (/30) subnet mask contains  $2^2 - 2 = 2$  host addresses You should not use the 255.255.248.0 subnet mask for the branch office. The branch office requires 4,000 host addresses, but a 255.255.248.0 subnet mask can accommodate only 2,046 host addresses. You should not use the 255.255.192.0 subnet mask for the branch office. A 255.255.192.0 subnet mask can accommodate 16,382 host addresses; however, your supervisor has instructed you to conserve IPv4 address space. You should not use the 255.255.128.0 subnet mask for the branch office. A 255.255.128.0 subnet mask can accommodate 32,766 host addresses; however, your supervisor has instructed you to conserve address space. You should not use the 255.255.0.0 subnet mask for the branch office. A 255.255.0.0 subnet mask can accommodate 65,534 host addresses; however, your supervisor has instructed you to conserve IPv4 address space.

### Original Exam C Question 43

[Topology note: RouterB already connects to the 192.168.0.0/25 network.]

Which of the following subnet addresses is most appropriate for the point-to-point link between RouterA and RouterB? (Select the best answer.)

- A. 192.168.0.0/25
- B. 192.168.0.192/26
- C. 192.168.0.128/30
- D. 192.168.0.128/25
- E. 192.168.0.130/30
- F. 192.168.0.0/30

## Correct Answer: C

### Explanation

The 192.168.0.128/30 subnet address is most appropriate for the point-to-point link between RouterA and RouterB. To conserve IP addresses, you should always use a /30 subnet mask for point-to-point links. A /30 subnet mask, which is equivalent to 255.255.255.252, indicates that 30 bits are used for the network portion of the address and that 2 bits remain for the host portion of the address, which allows for 2, or  $2^2 - 2$ , host addresses. The address 192.168.0.128 is the network address, 192.168.0.129 can be used for one end of the link, 192.168.0.130 can be used for the other end of the link, and 192.168.0.131 is the broadcast address. You should not use the 192.168.0.0/25 subnet address, because that subnet is already used by the network attached to the other interface of RouterB. The 192.168.0.0/25 subnet includes addresses from 192.168.0.0 through 192.168.0.127. A /25 subnet mask indicates that 25 bits are used for the network portion of the address and that 7 bits remain for the host portion of the address, which allows for 126, or  $2^7 - 2$ , usable host addresses. You should not use the 192.168.0.0/30 subnet address, because the addresses within this subnet are used by the network attached to the other interface of RouterB. The 192.168.0.0/30 subnet includes addresses from 192.168.0.0 through 192.168.0.4. You should not use the 192.168.0.128/25 subnet address, because that subnet range allows for 126, or  $2^7 - 2$ , host addresses from 192.168.0.129 through 192.168.0.254. Only two addresses are required for a point-to-point link, so using a /25 subnet mask would waste the other 124 addresses. You should not use the 192.168.0.130/30 subnet address, because the 192.168.0.130 IP address is not the start of a /30 subnet. Networks that are subnetted by using /30 masks are separated into groups of four addresses each; therefore, subnet addresses must be divisible by 4. The address 192.168.0.130 is a host address within the 192.168.0.128/30 subnet. You should not use the 192.168.0.192/26 subnet address, because that subnet range allows for 62, or  $2^6 - 2$ , host addresses from 192.168.0.193 through 192.168.0.254. Only two addresses are required for a point-to-point link, so using a /26 subnet mask would waste the other 60 addresses.

## Original Exam C Question 67

Given a host IP address of 48.25.24.71/21, what is the broadcast address for the subnetwork? (Select the best answer.)

- A. 48.25.24.240
- B. 48.25.25.254
- C. 48.25.31.254
- D. 48.25.31.255
- E. 48.25.24.255
- F. 48.25.24.127

## Correct Answer: D

### Explanation

The broadcast address for the subnetwork containing the host 48.25.24.71/21 is 48.25.31.255. An Internet Protocol (IP) address is composed of four groups of 8 binary bits, or 32 bits total. Each bit can store either a 1 or a 0 value. The address consists of two parts, a network portion and a host portion, which are divided by the use of a subnet mask. Like the IP address, the subnet mask is composed of four groups of 8 binary bits containing either a 1 or a 0 value. Because each group contains 8 bits of information, the groups are referred to as octets. Each octet ranges from 0 through 255 in decimal value. To determine the subnetwork address range of a given IP address/subnet mask combination, you must first identify the interesting octet within the subnet mask. The interesting octet is the first octet that contains a decimal value other than 255 or 0. The subnet mask in this example is /21. This notation is known as Classless Inter-Domain Routing (CIDR) notation. To calculate the network and host information for the network, you will need to convert the subnet mask to dotted decimal notation. To convert /21 from CIDR notation to dotted decimal notation, begin at the left and set the first 21 bits to a value of 1. Set the remaining 11 bits to 0. /21 = 11111111.11111111.1111000.00000000 Binary bit weight increases in significance from right to left, with the leftmost bit in each octet worth a decimal value of 128 and the rightmost bit worth a decimal value of 1. The decimal value for each octet is computed by adding up the bit weight for any bit containing a 1 within the octet. The following exhibit displays how to calculate the decimal value of the subnet mask octets based on the binary value assigned to each bit: ENN cu! S| DECIMAL EGE EE EE EE EE EE Eee OED EGE EE EE EE EE EE EEE OE! EGS EE EE EE EE EEE Eee OE T° Mo Bo oo Bo | /21 = 11111111.11111111.1111000.00000000 = 255.255.248.0 Now that the subnet mask is in dotted decimal notation, the interesting octet is easily identified, because the third octet has a decimal value of 248, not 0 or 255. When the interesting octet has been identified, the network numbers are determined by the weight of the least significant bit that is set to 1 in the interesting octet. Another way to determine the network numbers is to simply subtract the decimal value of the interesting octet from 256. The difference between 256 and 248 is 8; therefore, the networks will be arranged in multiples of 8. BIT WEIGHT Ee EE Ee EE Ee ES OE With the network numbers arranged by factors of 8, a partial list of the available networks using the 255.255.248.0 subnet mask in this scenario includes the following: 48.25.0.0 48.25.8.0 48.25.16.0 48.25.24.0 48.25.32.0 «and

so on The total number of hosts can be determined by the number of bits equal to 0 in the binary subnet mask. You can then calculate the number of hosts for a given subnetwork by using the formula  $2^n - 2$ , where n is the number of bits equal to 0 in the subnet mask. /21 = 11111111.11111111.11110000.00000000 There are 11 bits equal to 0 in the /21 subnet mask. Using the  $2^n - 2$  formula,  $2,048 - 2 = 2,046$ . Therefore, 2,046 hosts are available for each subnetwork when a subnet mask of /21 is applied. You must subtract 2 from the number of available hosts, because the first address is the subnetwork address and the last is the broadcast address. Likewise, you can determine the number of subnetworks available by examining the network mask. Because you are working with a Class A network, the first octet is excluded from the calculation. The formula to calculate the number of networks is  $2^n$ , where n is the number of bits equal to 1. XXXXXXXX.11111111.11110000.00000000 Excluding the first octet, 13 bits that equal 1 are still in the subnet mask. Using the # networks =  $2^n$  formula, you should find that 8,192 networks are available when a subnet mask of 255.255.248.0 is applied to a Class A network. Applying a subnet mask larger than the default subnet mask for a given network class is known as supernetting. This technique allows network engineers and administrators to combine and divide networks without regard to classful network boundaries, thus making more efficient use of IP address space. Based on the information above, an IP address of 48.25.24.71/21 is located in the 48.25.24.0 network. This network has an address range of 48.25.24.0 to 48.25.31.255. The first address in the subnetwork is 48.25.24.1, and the last usable host address is 48.25.31.254; thus, the broadcast address is 48.25.31.255.

## Days 16-18 - VLANs

Included Exam C items: Q42, Q60, Q65

### Original Exam C Question 42

issue the commands switch:

You following on a Catalyst 9300

SwitchA#configure terminal

SwitchA(config)#interface gigabitethernet 1/0/1

SwitchA(config-if)#switchport mode trunk

SwitchA(config-if)#switchport trunk encapsulation dot1q

SwitchA(config-if)#switchport trunk native vlan 44

SwitchA(config-if)#switchport trunk allowed vlan remove 55

Which of the following statements is true regarding VLAN traffic when it is sent over port Gi1/0/1? (Select the best answer.)

- A. traffic will be tagged.
- B. traffic will be untagged.
- C. All VLAN traffic will be untagged.
- D. VLAN 1 traffic will be untagged.
- E. VLAN 44 traffic will be untagged.

**Correct Answer: E**

Explanation

Traffic from virtual local area network (VLAN) 44 will be untagged when it is sent over port Gi1/0/1. VLAN 44 traffic is untagged because it has been configured as the native VLAN by the switchport trunk native vlan 44 command. By default, the native VLAN is VLAN 1. You can issue the switchport trunk native vlan vlan-id command to change the native VLAN. All VLAN traffic will not be untagged when it is sent over port Gi1/0/1; only native VLAN traffic will be untagged. Traffic from all other VLANs will be tagged when it is sent over port Gi1/0/1. A switch adds a VLAN tag to the header of each frame that belongs to a non-native VLAN before sending the frame out a trunked port so that the receiving device will know which VLAN the frame belongs to. Traffic from VLAN 1 will not be untagged when it is sent over port Gi1/0/1; it will be tagged because VLAN 1 is no longer the native VLAN. To reconfigure VLAN 1 to be the native VLAN, you can issue the switchport trunk native vlan 1 command or the no switchport trunk native vlan command. Traffic from VLAN 55 cannot be sent over port Gi1/0/1. The switchport trunk allowed vlan remove 55 command removes VLAN 55 from the list of allowed VLANs that can be trunked over port Gi1/0/1. The switchport trunk allowed vlan {add | all | except | remove} vlan-list command is issued from interface configuration mode to manually prune VLANs. Manual pruning enables an administrator to which allowed denied trunk strictly specify VLANs are or on a port. You can issue the show interfaces trunk command to display the list of ports that are configured for trunking, the native VLAN for each trunk port, and the list of currently allowed VLANs for each trunk port. The following displays the output of the show interfaces trunk command: SwitchA#show interfaces trunk Port Mode Encapsulation Status Native vlan Gi1/0/1 desirable

802.1q trunking 44 Vlans allowed trunk Port on Gi0/0/1 1-54,56-4994 Vlans allowed and active in domain Port management Gi0/0/1 1-10,44 Vlans in and Port spanning tree forwarding state not pruned Gi0/0/1 1-10,44

### Original Exam C Question 60

SwitchB Er: VLAN 111 SwitchA and SwitchB are connected over an 802.1Q trunk link. The native VLAN for the trunk link is configured as VLAN 11 on SwitchA and VLAN 111 on SwitchB. All of the hosts reside in the same IP subnet.

What of the following is true regarding the connectivity between the hosts in this scenario? (Select the best answer.)

- A. HostC can ping HostD. O HostB can ping HostC.
- C. None of the hosts can ping each other.
- E. HostA can ping HostB.

**Correct Answer: D**

Explanation

In this scenario, HostD can ping HostA. One virtual local area network (VLAN) is considered the native VLAN when 802.1Q encapsulation is configured on a trunk link. Traffic over the native VLAN is not that header is not added the frame. When tagged; is, an 802.1Q to a switch receives a frame without the switch knows that the frame is part of the native an 802.1Q header, VLAN. By default, VLAN 1 is identified as the native VLAN. In this scenario, the native VLAN for the trunk link on SwitchB is configured as VLAN 111. When HostD sends a ping message to HostA, SwitchB recognizes that HostD is a member of VLAN 111. Therefore, the traffic from HostD is sent untagged across the trunk link to SwitchA. When SwitchA receives the untagged frames from SwitchB, it forwards them to the native VLAN configured for its side of the link, which in this scenario is VLAN 11. Because HostA is a member of VLAN 11, it will receive the ping messages from HostD and will be able to reply by using the same process in reverse. HostB cannot ping HostC. When HostB attempts to send a ping message to HostC, SwitchA will recognize that HostB is a member of VLAN 111 and will add an 802.1Q header to tag the traffic as it crosses the trunk link. However, when SwitchB receives the tagged traffic from HostB, it will recognize the in the header its native traffic from the VLAN ID 802.1Q as locally configured VLAN. Because native VLAN should not be tagged on the trunk link by default, SwitchB will consider the traffic from HostB to be erroneous and will discard it. HostA cannot ping HostB and HostC cannot ping HostD because each pair resides in different VLANs on the same switch. In addition, because all hosts in this scenario are configured to reside in the same Internet Protocol (IP) subnet, neither switch could be configured to route traffic between its different VLANs even if the switches were capable of Layer 3 forwarding.

### Original Exam C Question 65

issue the commands You following on a Cisco switch:

```
SwitchA#configure terminal
SwitchA(config)#interface fastethernet 0/5
SwitchA(config-if)#switchport mode trunk
SwitchA(config-if)#switchport trunk encapsulation dot1q
SwitchA(config-if)#switchport trunk allowed vlan add 1-99
SwitchA(config-if)#switchport trunk allowed vlan remove
SwitchA(config-if)#switchport trunk allowed vlan except
```

Which VLANs are allowed to be trunked over port FastEthernet 0/5? (Select the best answer.)

- A. VLANs 1-99 and 200-299
- B. VLANs 200-299
- C. VLANs 1-199 and 300-4094
- D. VLANs 1-99 and 200-4094
- E. VLANs 1-99
- F. VLANs 1-99 and 300-4094

**Correct Answer: C**

Explanation

Virtual local area networks (VLANs) 1-199 and 300-4094 are allowed to be trunked over port FastEthernet 0/5. When you create a trunk port by issuing the switchport mode trunk command, all VLANs are allowed over the trunk by default. The switchport trunk allowed vlan command is used to perform manual pruning, which is used to allow traffic only from specified VLANs over the trunk. The syntax of the switchport trunk allowed vlan command is switchport trunk allowed vlan {add | all | except | remove} vlan-list. The add keyword is used to add VLANs to the list of allowed VLANs. The all keyword is used to

restore the default behavior of the switch, which allows all VLANs over the trunk. The `except` keyword is used to add all VLANs to the list of allowed VLANs except those specified in the `vlan-list` parameter. The `remove` keyword is used to remove VLANs from the list of currently allowed VLANs. If a keyword is not specified, the list of allowed VLANs will be replaced by the in the command. The `vlan-list` be list of VLANs specified parameter can a single VLAN, a VLANs separated by commas, or a range of VLANs separated by a dash. The `switchport trunk allowed vlan add 1-99` command adds VLANs 1-99 to the list of allowed VLANs for the trunk. However, in this scenario, the `switchport trunk allowed vlan add 1-99` command is unnecessary, because all VLANs are allowed over a trunk port by default. The `switchport trunk allowed vlan remove 100-199` command removes VLANs 100-199 from the list of allowed VLANs for the trunk. Therefore, only VLANs 1-99 and 200-4094 are allowed after you issue the `switchport trunk allowed vlan remove 100-199` command on your switch. The `switchport trunk allowed vlan except 200-299` command adds all VLANs to the list of allowed VLANs on the trunk except for VLANs 200-299. Unlike the `remove` keyword, which modifies the existing list of allowed VLANs, the `except` keyword replaces the list of allowed VLANs. Therefore, VLANs 1-199 and 300-4094 are allowed after you issue the `switchport trunk allowed vlan except 200-299` command. To allow only VLANs 1-99 on a newly created trunk, you could issue the `switchport trunk allowed vlan 1-99` command. To allow only VLANs 200-299 on a newly created trunk, you could issue the `switchport trunk allowed vlan 200-299` command. Similarly, to allow only VLANs 1-99 and 200-299 on a newly created trunk, you could issue the `switchport trunk allowed vlan 1-99` and `switchport trunk allowed vlan add 200-299` commands. To allow only VLANs 1-99 and 300-4094 on a newly created trunk, you could issue the `switchport trunk allowed vlan remove 100-199` command. Similarly, to allow only VLANs 1-99 and 200-4094 on a newly created trunk, you could issue the `switchport trunk allowed vlan remove 100-199` command. You can use the `show interfaces trunk` command to display the list of ports that are configured for trunking and the list of currently allowed VLANs for each trunk port. The command is also useful for troubleshooting native VLAN mismatches. A native VLAN mismatch can cause traffic from one VLAN on a switch to be sent to a different, incorrect VLAN on another switch. By comparing the output of the `show interfaces trunk` command on one switch with the output of the `show interfaces trunk` command on the connected switch, you can determine whether a native VLAN mismatch exists between the two switches. The following displays the output of the `show interfaces trunk` command: SwitchA#show interfaces trunk Port Mode Encapsulation Status Native vlan Fa0/5 desirable 802.1q trunking 1 Port Vlans allowed on trunk Fa0/5 1-199,30e-4094 Port Vlans allowed and active in management domain Fae/S5 1-9 Port Vlans in spanning tree forwarding state and not pruned Fae0/5 1-9 The `show interfaces switchport` command can also be used to detect native VLAN mismatches.

## Days 20-21 - Spanning Tree Protocol (STP)

Included Exam C items: Q3, Q10, Q39, Q44

### Original Exam C Question 3

Your company's network is shown in the topology diagram. PC1 is connected to FastEthernet 0/5 on ASW1. PC2 is connected to FastEthernet 0/5 on ASW2.

Configure STP as follows:

- Configure the ports connected to PC1 and PC2 to bypass the STP convergence process and immediately transition to the forwarding state.
- Configure DSW1 as the root bridge for VLAN 5 and the secondary bridge for VLAN 6.
- Configure DSW2 as the root bridge for VLAN 6 and the secondary bridge for VLAN 5.

**Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam C PDF.**

### Original Exam C Question 10

You issue the `show log` command to determine the cause of an err-disabled port and notice the following lines of output:

%SPANTREE-SP-2-BLOCK\_BPDUGUARD:

Received BPDU on port FastEthernet0/11 with BPDU Guard enabled. Disabling port.

%PM-SP-4-ERR\_DISABLE:

bpduguard error detected on Fa0/11, putting Fa0/11 in err-disable state The err-disabled recovery feature is enabled for the BPDU-guard-error cause with its default settings.

What is the maximum amount of time that can pass before the err-disabled port can attempt recovery?

(Select the best answer.)

B. 30seconds

- C. 600seconds
- D. 60seconds

**Correct Answer: A**

#### Explanation

A maximum of 300 seconds can pass before the err-disabled port in this scenario can attempt recovery. The err-disabled recovery feature enables a switch to automatically re-enable an err-disabled port after a specified amount of time has passed. The err-disabled recovery feature can be configured for all causes of err-disabled ports or only for specific causes. In addition, the time interval before a port can attempt recovery is configurable and defaults to 300 seconds. When in err-disabled issue the show command determine that a port an state, you can log to cause. However, if there are a large number of log entries, it may be more expedient to issue the show errdisable recovery command. The show errdisable recovery command provides a summary of configured err-disabled status reasons, err-disabled ports, and the time remaining before each port can attempt recovery. The following sample output from the show errdisable recovery command from this scenario indicates that the FastEthernet 0/11 interface has 284 seconds remaining before it can attempt recovery: ErrDisable Reason Timer Status uddl Disabled bpduguard Enabled security-violatio Disabled channel-misconfig Disabled vps Disabled pagp-flap Disabled dtp-flap Disabled link-flap Disabled 12ptguard Disabled psecure-violation Disabled gbic-invalid Disabled dhcp-rate-limit Disabled Disabled unicast-flood Enabled storm-control arp-inspection Enabled Timer interval: 300 seconds Interfaces that will be enabled at the next timeout: Interface Errdisable reason Time left(sec) Fae0/11 bpduguard 284

### Original Exam C Question 39

You issue the spanning-tree guard root command on a switch port that you are connecting to a new, unconfigured switch. Which of the following are you most likely attempting to do? (Select the best answer.)

- A. prevent loops on a port that could erroneously receive BPDUs
- B. prevent a port from transitioning through all of the STP states
- C. prevent the new switch from being elected root
- D. prevent loops from an interruption of BPDU flow

**Correct Answer: C**

#### Explanation

Most likely, you are attempting to prevent the new switch from being elected root if you issue the command spanning-tree guard root on a switch port that you are connecting to a new, unconfigured switch. The spanning-tree guard root command configures the Spanning Tree Protocol (STP) root guard feature. Root guard is used to prevent newly introduced switches from being elected as the new root switch. This allows administrators to maintain control over which switch is the root. When STP is used, the device with the lowest switch priority is elected the root. If a new device is added to the network with a lower priority than the current root, it will become the new root. However, this could cause the network to reconfigure in unintended ways. To prevent this, root guard can be applied. Root guard is applied on a per-port basis by issuing the spanning-tree guard root command. If root guard is enabled on a loop guard-enabled port, loop guard will be automatically disabled. You are not attempting to prevent a port from transitioning through all of the STP states. If you want to ensure that a port immediately transitions to the STP forwarding state, you should enable PortFast on the port. PortFast is a feature that provides immediate accessibility to the network for edge ports, such as access ports that are connected to end-user workstations. PortFast transitions the port into the STP forwarding state without going through the STP listening and learning states. Because the ports are not expected to receive bridge protocol data units (BPDUs), they are not required to listen for BPDUs and learn the network topology. It is important to note that if PortFast is enabled on a port that is connected to another switch, the potential for creating spanning tree loops significantly increases. You are not attempting to prevent loops from an interruption of BPDU flow. The loop guard feature prevents nondesignated ports from inadvertently forming bridging loops if the steady flow of BPDUs is interrupted. When the port stops receiving BPDUs, loop guard puts the port into the loop- inconsistent state, which keeps the port in a blocking state. After the port starts receiving BPDUs again, loop guard automatically re-enables the port so that it transitions through the normal STP states. You can enable loop guard for the entire switch by issuing the spanning-tree loopguard default command in global configuration mode, or you can enable loop guard for specific ports by issuing the spanning-tree guard loop command in interface configuration mode. You are not attempting to prevent loops on a port that could erroneously receive BPDUs. BPDU guard is used to disable ports that erroneously receive BPDUs. BPDU guard is typically applied to edge ports that have PortFast enabled. Because PortFast automatically places ports into a forwarding state, a switch that has been connected to a PortFast-enabled port could cause switching loops. However, when is the of will result in the into BPDU guard applied, receipt a BPDU on a port port being placed the err-disabled state, which prevents loops from occurring. BPDU guard should be enabled on ports that have been enabled with PortFast so that BPDU guard can prevent a rogue switch from

modifying the When such receives that into STP topology. a port a BPDU, BPDU guard immediately puts port the err-disabled state and shuts down the port. The port must then be manually re-enabled, or it can be recovered automatically by configuring the errdisable recovery cause bpduguard command and the errdisable recovery interval interval command.

### Original Exam C Question 44

Priority: 32768 MAC: 0000.0c65.4ba6 SwitchA SwitchB Priority: 32768 MAC: 0000.0c60.0717 You need to determine how traffic will flow between the two switches in the topology shown above. No VLANs are configured on the switches.

Which port will be blocked? (Select the best answer.)

- A. The FastEthernet port on SwitchA will be blocked.
- B. No ports will be blocked.
- C. The FastEthernet port on SwitchB will be blocked.
- D. The GigabitEthernet port on SwitchA will be blocked.
- E. The GigabitEthernet port on SwitchB will be blocked.

**Correct Answer: A**

#### Explanation

The FastEthernet port on SwitchA will be blocked. When two redundant links exist between two switches, Spanning Tree Protocol (STP) will prevent a switching loop by transitioning one of the ports to the blocking state. The blocking state in STP is equivalent to the discarding state in Rapid STP (RSTP). Neither STP nor RSTP supports load balancing over redundant links. To identify which port will be blocked, you must first identify which switch will become the root bridge. The root bridge is the switch with the lowest bridge ID (BID). The BID is composed of a 2-byte bridge priority and a 6-byte Media Access Control (MAC) address. Because the bridge priority is the first part of the BID, a switch with the lowest bridge priority will have the lowest BID. The bridge priority can be set by issuing the spanning-tree priority value command, where value is a multiple of 4096 in the range from 0 through 61440. In this scenario, SwitchA and SwitchB are both set to the default priority of 32768. When two or more switches have the lowest priority on the switched network, the switch with the lowest MAC address becomes the root bridge; in MAC addresses, numbers are lower than letters and the hexadecimal value A is lower than the hexadecimal value F. Because SwitchB has a lower MAC address than SwitchA, SwitchB will become the root bridge. Electing a root bridge is one of the first steps in the STP process. STP also optimizes the network by finding the lowest cost paths, determines which port becomes the root port of each switch, and determines which ports become designated ports. STP determines port states through the use of the Spanning Tree Algorithm (STA). The STA uses the following procedure to determine port state: A root switch is elected, and each active port on the root switch is placed into the forwarding state. e Each nonroot switch places the port with the lowest-cost path to the root switch, which is known as the root port, into the forwarding state. e Ports with the lowest cost to nonroot switches, which are known as designated ports, are also placed into the forwarding state. All other ports are placed into the blocking state. Ports on a root bridge are never blocked. Therefore, neither of the ports on SwitchB will be blocked. To determine which of the ports on SwitchA must be blocked, SwitchA will identify the port that has the lowest path cost to the root bridge. The higher the link speed, the lower the path cost. Therefore, the FastEthernet port on SwitchA will be blocked because it has a lower link speed than the GigabitEthernet port. If both ports had the same link speed, the port that received the bridge protocol data unit (BPDU) with the lowest sending BID would become the root port and would transition to the forwarding state. When multiple ports are connected to the same switch, the port that received the BPDU with the lowest sending port ID would become the root port and would transition to the forwarding state; the other ports would transition to the discarding state. The port ID consists of the port priority and the port number. The port configured with the lowest port priority would also have the lowest port ID; port numbers are considered only when port priorities are equal.

## Day 23 - EtherChannel

Included Exam C items: Q50

### Original Exam C Question 50

Select the EtherChannel operating modes and match the SwitchA/SwitchB mode combinations that result in a working EtherChannel. Modes may be used more than once.

Modes: off, auto, desirable, passive, active, on.

### Correct Answer: Valid EtherChannel mode pairings:

- on + on
  - PAgP: desirable + desirable, desirable + auto, auto + desirable
  - LACP: active + active, active + passive, passive + active
- Invalid: auto + auto, passive + passive, or mixing PAgP/LACP modes.

### Explanation

EtherChannel is used to bundle two or more identical, physical interfaces into a single logical link between switches. An EtherChannel can be permanently established between switches, or it can be negotiated by using one of two aggregation protocols: the Cisco-proprietary Port Aggregation Protocol (PAgP) or the open-standard Institute of Electrical and Electronics Engineers (IEEE) 802.3ad protocol, which is also known as Link Aggregation Control Protocol (LACP). The EtherChannel aggregation protocol must match on each switch, or they will be unable to dynamically establish an EtherChannel link between them. In addition, the channel groups on each switch must operate in compatible modes in order to create a functional EtherChannel link. The channel-group number mode {on | active | passive | {auto | desirable} [non-silent]} command is used to configure the operating mode for an interface, or range of interfaces, in a channel group. The following table displays the channel-group modes that will result in a valid EtherChannel configuration:

| SwitchA   | SwitchB | Result |
|-----------|---------|--------|
| desirable | off     | auto   |
| passive   | active  | on     |
| active    | on      | on     |

The on keyword configures the channel group to unconditionally create the channel with no LACP or PAgP negotiation. In the on mode, a functional EtherChannel exists only if a channel group that is in the on mode is connected to another channel group that is also in the on mode. If either side of the link is not in the on mode, packet loss can occur. In addition, while in on mode, the channel group ignores any LACP or PAgP configuration information. The active and passive keywords can be used only with LACP. The active keyword configures the channel group to actively negotiate LACP, and the passive keyword configures the channel group to listen for LACP negotiation to be offered. Either or both sides of the link must be set to active to establish an EtherChannel over LACP; setting both sides to passive will not establish an EtherChannel over LACP. The auto, desirable, and non-silent keywords can be used only with PAgP. The desirable keyword configures the channel group to actively negotiate PAgP, and the auto keyword configures the channel group to listen for PAgP negotiation to be offered. Either or both sides of the link must be set to desirable to establish an EtherChannel over PAgP; setting both sides to auto will not establish an EtherChannel over PAgP. The optional non-silent keyword requires that a port receive PAgP packets before the port is added to the channel. You can issue the show etherchannel summary command to verify the status of an EtherChannel link and to determine which aggregation protocol, if any, was used to establish the link. The following sample output indicates that the PAgP protocol was used to successfully establish an EtherChannel link:

```
SwitchA#show etherchannel summary
Flags: D - down P - bundled in port-channel s - suspended - Hot-standby (LACP only)
L - Layer3 S - Layer2 1 in use % - failed to allocate aggregator BC - not in use, minimum links not met c - unsuitable for bundling
- - waiting to be aggregated d - default port
Number of channel-groups in use: 1 Number of aggregators: 1
Group Port-channel Protocol Ports
1 Po1(US) PAgP Fa0/5(P) Fae/6(P)
```

## Days 26-28 - OSPF

Included Exam C items: Q2, Q11, Q19, Q28, Q47, Q63

### Original Exam C Question 2

You are the network administrator for your company's Nashville and Memphis area branches. Burbank area users are reporting that they are not able to access some shared file server resources in Memphis. In addition, users from all branches of your company are reporting that they cannot access the Internet.

You have been asked to verify that the OSPF configurations in Nashville and Memphis meet your company's requirements. Any configuration decisions should match the topology and require a minimum amount of administrative effort. Repair any configuration errors that prevent PC1 from pinging the MEM router's Loopback 0 interface and that prevent the rest of the company from accessing the Internet. The ISP router's Serial 0/0 interface has been assigned 203.0.113.1.

### Requirements:

- All OSPF routers must be single-area, nonbackbone routers, except BNA and HQ.
- All OSPF routers must advertise the IP address assigned to Loopback 0.
- All OSPF routers must be manually configured with a router ID matching the Loopback 0 address.
- All OSPF routers must advertise each physical interface's network address, not its IP address.

- All OSPF routers must be able to access the Internet by way of HQ.

Make only the changes required to accomplish the task. You do not have access to BUR, ISP, or TPA.

**Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam C PDF.**

### Original Exam C Question 11

and advertise all available interfaces. The serial interfaces You configure OSPF on a router are configured with PPP encapsulation and are using the default OSPF network type and timer values.

Which of the following statements is correct regarding how OSPF operates on the serial interfaces? (Select the best answer.)

- A. DR and BDR elections are performed.
- B. The neighbor command is required to establish adjacencies.
- C. Unicast updates are sent.
- D. The hello timer is set to 10 seconds, and the dead timer is set to 40 seconds.

**Correct Answer: D**

#### Explanation

The hello timer is set to 10 seconds and the dead timer is set to 40 seconds. The serial interfaces are configured with Point-to-Point Protocol (PPP) encapsulation, which uses the Open Shortest Path First (OSPF) point-to-point network type by default. There are five OSPF network types: © Broadcast © Nonbroadcast © Point-to-point © Point-to-multipoint broadcast © Point-to-multipoint nonbroadcast On point-to-point networks, designated router (DR) and backup designated router (BDR) elections are not performed. Multicast updates are sent, so manual configuration of neighbor routers with the neighbor command is not required. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF point-to-point network, you should issue the ip ospf network point-to-point command. The OSPF point-to-point network type is enabled by default on High-Level Data Link Control (HDLC) and Point-to-Point Protocol (PPP) serial interfaces. On broadcast networks, DR and BDR elections are performed. Multicast updates are sent, so the manual configuration of neighbor routers with the neighbor command is not required. By default, the hello timer is set to 10 seconds and the dead timer is set to 40 seconds. To configure an OSPF broadcast network, you should issue the ip ospf network broadcast command. The OSPF broadcast network type is enabled by default on Fiber Distributed Data Interface (FDDI) and Ethernet interfaces, including Fast Ethernet and Gigabit Ethernet interfaces. On nonbroadcast networks, DR and BDR elections are performed. Nonbroadcast networks do not allow multicasts; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. By default, the hello timer is set to 30 seconds and the dead timer is set to 120 seconds. To configure an OSPF nonbroadcast network, which is also called a nonbroadcast multiaccess (NBMA) network, you should issue the ip ospf network non-broadcast command. The OSPF nonbroadcast network type is enabled by default on Frame Relay and X.25 interfaces. broadcast networks OSPF point-to-multipoint operate just like OSPF point-to-point networks except the hello timer is set to 30 seconds and the dead timer is set to 120 seconds by default. To configure an OSPF point-to-multipoint broadcast network, you should issue the ip ospf network point-to-multipoint broadcast command. OSPF point-to-multipoint nonbroadcast networks operate just like OSPF point-to-multipoint broadcast networks except that multicasts cannot be sent; therefore, manual configuration of neighbor routers with the neighbor command is required so that OSPF sends unicast updates. To configure an OSPF point-to-multipoint nonbroadcast network, you should issue the ip ospf network point-to-multipoint non-broadcast command.

### Original Exam C Question 19

4 RouterA Router!

You administer the OSPF network in the above exhibit.

RouterA has been configured with a default route to the ISP through the Serial 0/0 interface. RouterB is not configured with a gateway of last resort. OSPF has been properly configured on both routers and is converged. You want RouterB to forward traffic that is destined for the Internet to RouterA.

Which of the following commands could you issue on RouterA to accomplish your goal? (Select the best answer.)

- A. RouterA(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
- B. RouterA(config-router)#default-information originate
- C. RouterA(config)#ip route 0.0.0.0 0.0.0.0 serial 0/28
- D. 10.10.10.2 0.0.0.3 area 0
- E. RouterA(config)#ip route 0.0.0.0 0.0.0.0 fastethernet 1/0

**Correct Answer: B**

## Explanation

You could issue the following command to configure RouterB with a gateway of last resort that would cause RouterB to forward traffic that is destined for the Internet to RouterA: RouterA(config-router)#default-information originate By default, Open Shortest Path First (OSPF) does not advertise a default route to neighboring routers. The default-information originate command configures an OSPF router to inject its default route into OSPF as an external route, thereby advertising its default route to neighboring routers and causing the neighbor routers to become configured with a gateway of last resort. You should not issue the ip route 0.0.0.0 0.0.0.0 fastethernet 1/0 command or the ip route 0.0.0.0 0.0.0.0 10.10.10.2 command on RouterA. Both of those commands would configure RouterA to use RouterB's FastEthernet 1/0 interface as a gateway of last resort. You do not need to issue the ip route 0.0.0.0 0.0.0.0 serial 0/0 command on RouterA. In this scenario, RouterA has already been configured with a default route to the Internet service provider (ISP) through the Serial 0/0 interface. You do not need to issue the network 10.10.10.2 0.0.0.3 area 0 command on RouterA. In this scenario, OSPF has already been properly configured on RouterA and RouterB and is fully converged. Therefore, all network commands have already been issued.

## Original Exam C Question 28

\_RouterC\_ > ae RouterE You administer the network in the above exhibit. OSPF is configured with the default settings. You issue the show ip route ospf command on RouterE.

Which of the following are you least likely to see in the output? (Select the best answer.)

- A. o 192.0.6.1 [110/1] via 10.10.10.7, 00:23:40, FastEthernet1/1
- B. o 192.0.5.1 [110/65] via 10.10.10.10, 00:23:40, Serial0/0
- C. o 192.0.3.1 [110/66] via 10.10.10.7, 00:23:40, FastEthernet1/0
- D. o 192.0.4.1 [110/2] via 10.10.10.7, 00:23:40, FastEthernet1/0

**Correct Answer: A**

## Explanation

Of the choices provided, you are least likely to see the following output from the show ip route command on RouterE: 192.0.6.1 via FastEthernet1/1 i) 10.10.10.7, 00:23:40, The 0 in the output above indicates that the path to 192.0.6.1 was inserted into the routing table by Open Shortest Path First (OSPF). However, the 192.0.6.0 network is directly connected to RouterE in the topology in this scenario. By default, a router prefers a directly connected path to a path learned by a routing protocol. Therefore, the path to 192.0.6.1 would not have been inserted into the routing table by OSPF. When the show ip route command is issued with a routing protocol keyword, the resulting output displays only the routes that were inserted into the routing table by the specified routing protocol. In this scenario, the show ip route ospf command would display only the routes that were inserted into the routing table on RouterE by OSPF. If you were to issue the show ip route ospf command on RouterE in this scenario, you would most likely see a total of six paths in the output that were learned by RouterE. In addition to the 192.0.3.0, 192.0.4.0, and 192.0.5.0 networks, all of which are directly connected to other OSPF routers in the topology, RouterE would use OSPF to learn about the networks that are configured on the interfaces that directly connect RouterB to RouterC, RouterB to RouterD, and RouterC to RouterD. If RouterB were connected to networks outside the OSPF topology in this scenario and were configured to redistribute those networks into OSPF, RouterE would additionally learn about the routes beyond RouterB by using OSPF. Similarly, RouterE could learn about a default route through RouterB by using OSPF if the default-information originate command were issued on RouterB.

## Original Exam C Question 47

OSPF Area 0 2 RouterA 192.0.5.0/24 192.0.6.0/24 192.0.4.0/24 192.0.3.0/24 You issue the show ip route command on RouterB and receive the following output:

<output omitted> 10.0.0.0/8 is variably subnetted, 6 subnets 90 10.1.1.0/30 [110/65] via 10.1.1.14, FastEthernet1/0 10.1.1.8/30 [110/65] via 10.1.1.14, FastEthernet1/e0 10.1.1.12/30 is directly connected, FastEthernet1/e0 10.1.1.13/32 is directly connected, 10.1.1.16/38 is directly connected, Serial0/0 10.1.1.17/32 is directly connected, Serial0/0 192.0.6.0/24 is variably subnetted, 2 subnets 192.0.6.0/24 is directly connected, FastEthernet2/e 192.0.6.1/32 is directly connected, FastEthernet2/0 192.0.5.0 [110/65] via 10.1.1.18, 00:02:04, Serial0/0 192.0.3.0 [110/66] via 10.1.1.14, 00:01:29, FastEthernet1/e 192.0.4.0 [110/2] via 10.1.1.14, 00:01:29, FastEthernet1/0 Which of the following statements are accurate about the route to 192.0.3.0/24? (Select 2 choices.)

- A. The metric is 110.
- B. The bandwidth of the link is 66 Kbps.
- C. The AD is 66.
- F. The bandwidth of the link is 110 Kbps.

## Correct Answer: D, E

### Explanation

The administrative distance (AD) of the route in the scenario is 110, and the metric is 66. In the output of the show ip route command, the first number within the brackets indicates the AD. When multiple different routes to a network exist and each route uses a routing protocol, a router prefers routing protocol with the lowest AD. The following list contains the most commonly used ADs: Route Source Connected route Static route EIGRP summary route eBGP Internal EIGRP IGRP OSPF IS-IS RIP External EIGRP iBGP Unknown The AD of the route in this scenario is 110, the same as the default distance for an Open Shortest Path First (OSPF) route. ADs for a routing protocol can be manually configured by issuing the distance command in router configuration mode. For example, to change the AD of OSPF process ID 1 from 110 to 80, you could issue the following commands: RouterB(config)#router ospf 1 RouterB(config-router)#distance 80 The second number within the brackets indicates the metric. Because this is an OSPF link, as indicated by the 0 at the start of the route statement, the second number within the brackets indicates the cost metric. Therefore, the cost of the link in this scenario is 66. When two OSPF paths exist to the same destination, the router will choose the OSPF path with the lowest cost. OSPF calculates cost based on the bandwidth of an interface: the higher the bandwidth, the lower the cost. To calculate the cost, divide 100,000,000 by the bandwidth in bits per second. Thus a 100-Mbps link would have a cost of 1, a 10-Mbps link would have a cost of 10, a T1 link would have a cost of 64 (100,000,000 / 1,544,000), and a 64-Kbps link would have a cost of 1,562. As the hops between a router and a destination increase, the cost increases by the bandwidth calculation of the additional links. In this scenario, the cost of the FastEthernet link from RouterB to RouterC is 1 Kbps. The serial connection between RouterC and RouterD adds the cost of a T1 link, which is 64 Kbps. Finally, the FastEthernet link between RouterD and the 192.0.3.0/24 network adds a cost of 1 Kbps. Therefore, the cost for RouterB to reach the 192.0.3.0/24 network is 1 + 64 + 1, or 66.

## Original Exam C Question 63

RouterA and RouterB are OSPF routers that are configured with matching hello timer values, dead timer values, and authentication passwords. Both routers are operating in OSPF Area 0 and have been configured to operate on the same subnet. However, the routers cannot communicate.

Which of the following could be the problem? (Select 2 choices.)

- A. a passive interface configuration
- B. mismatched cost values
- D. mismatched router IDs
- E. a mismatched process ID

## Correct Answer: A, C

### Explanation

Mismatched maximum transmission unit (MTU) values could prevent communication between RouterA and RouterB. Alternatively, a passive interface configuration could prevent communication between the two Open Shortest Path First (OSPF) routers. The routers will be stuck in Database Description (DBD) exchange after RouterA exchanges OSPF hello packets with RouterB. OSPF interfaces with mismatched MTU values will not be able to form an adjacency if the MTU mismatch is detected during the DBD packet exchange. You can configure OSPF to ignore MTUs or manually configure matching MTUs on both sides of a link to solve this problem. To configure OSPF to ignore MTU mismatches, you should issue the ip ospf mtu-ignore command in OSPF router configuration mode. Neighbor relationships cannot be established over interfaces that have been configured with the passive-interface command. You can determine whether the passive-interface command has been configured on an interface by issuing the show running-config command or by issuing the show ip ospf interface or the show ip protocols command. The show ipv6 protocols command is the OSPF version 3 (OSPFv3) equivalent of the show ip protocols command. Mismatched cost values will not prevent OSPF routers from communicating. OSPF uses link cost as a metric to determine the best route to a destination. The best route is the route with the lowest link cost. By default, the link cost is determined by the bandwidth of the link. The higher the bandwidth, the lower the cost. If multiple OSPF routes exist to a destination and the cost of each route is equal, a Cisco router will load balance traffic equally between those routes. A mismatched process ID will not prevent OSPF routers from communicating. The process ID is locally significant to the router. To enable OSPF on a router, you should issue the router ospf process-id command from global configuration mode, where process-id is any positive integer in the range from 1 through 65,535 that identifies the routing process. A mismatched router ID will not prevent OSPF routers from communicating. However, using the same router ID on more than one OSPF router will cause communication problems. The OSPF router ID is used to uniquely identify each router on the network. You can manually configure a router ID for a router by issuing the router-id ip-address command in router configuration mode, where ip-address is the address that you want OSPF to acknowledge as the new router ID. If you manually configure a router ID, it is important to ensure that the router ID is unique to a given OSPF router on the network. Two OSPF routers that have the same router ID will

not be able to form an adjacency. If you do not manually configure a router ID, the router ID is the highest Internet Protocol (IP) address configured among loopback interfaces on the router. If no loopback addresses are configured on the router, the router ID is the highest IP address configured among all physical interfaces on the router.

## Day 29 - First Hop Redundancy Protocols

Included Exam C items: Q18, Q72

### Original Exam C Question 18

Which of the following is a Cisco-proprietary FHRP that elects an AVG and up to four primary AVFs? (Select the best answer.)

- A. VRRP
- B. LACP
- C. GLBP
- D. HSRP

**Correct Answer: C**

#### Explanation

Gateway Load Balancing Protocol (GLBP) is a Cisco-proprietary First- Hop Redundancy Protocol (FHRP) that elects an active virtual gateway (AVG) and up to four primary active virtual forwarders (AVFs). FHRPs are protocols that are used to provide Layer 3 gateway redundancy, such as failover and load balancing. Providing Layer 3 redundancy ensures that hosts on a local area network (LAN) will have a backup path to external networks should a primary path fail or become too congested to forward traffic. Layer 3 devices in an FHRP configuration typically share a virtual Internet Protocol (IP) address that is then configured as the default gateway on each host for which the device is to forward traffic. The FHRP devices might also share a single virtual Media Access Control (MAC) address or provide multiple virtual MAC addresses, depending on the protocol. FHRPs typically use a priority system to elect a primary Layer 3 forwarding device, which is known as an AVG, an active router, or a master router, depending on the protocol. The same priority system elects either a single or multiple backup- forwarding devices. Each GLBP group contains an AVG that is elected based on which router is configured with the highest priority value or the highest IP address value if multiple routers are configured with the highest priority value. The other routers in the GLBP group are configured as primary or secondary AVFs. GLBP can support up to 1,024 virtual routers on a physical interface. The AVG in a GLBP group assigns a virtual MAC address to a maximum of four primary AVFs; all other routers in the group are considered secondary AVFs and are placed in the listen state. When the AVG receives Address Resolution Protocol (ARP) requests that are sent to the virtual IP address for the GLBP group, the AVG responds with different virtual MAC addresses. This provides load balancing, because each of the will of the traffic the virtual IP address. The primary AVFs participate by forwarding a portion sent to primary difference between GLBP and other FHRPs is that, by default, GLBP load balances between every router in the GLBP group. Other protocols either cannot load balance or require additional configuration in order to load balance. Hot Standby Router Protocol (HSRP) is a Cisco-proprietary FHRP; however, HSRP elects only an active router and a standby router. Based on priority value, HSRP elects a single active router and a standby router. The active router is the router with the highest priority; it forwards packets, responds to ARP requests with a virtual MAC address, and can be the only router that is explicitly configured with the virtual IP address. The standby router is the router with the second-highest priority. If multiple HSRP routers have the same priority, the router with the highest IP address is elected as the active router. The router with the second-highest IP address is elected as the standby router, which will assume the role of the active router if the active router fails. To participate in the active and standby router election process, each HSRP router must be a member of the same group. An HSRP group is identified by a group number from 0 through 255. The default HSRP group value is 0. Unlike GLBP, HSRP does not load balance by default. However, it is possible to load balance traffic between HSRP gateways by creating up to 255 HSRP groups on an interface and configuring each group so that it elects a different active router. This is known as a multigroup HSRP configuration. Virtual Router Redundancy Protocol (VRRP) is not a Cisco-proprietary FHRP. VRRP is an Internet Engineering Task Force (IETF)-standard FHRP that is supported by both Cisco and non-Cisco devices. Similar to HSRP, VRRP elects a master router that forwards packets, responds to ARP requests, and can be the only router that is explicitly configured with the virtual IP address. It is important to note that Cisco-specific enhancements to VRRP might not be available when connecting a Cisco device that is using VRRP to a non-Cisco device that is using VRRP. If only Cisco devices are used in the topology and a choice between HSRP and VRRP is available, Cisco recommends using HSRP. Both HSRP and VRRP can be used to configure failover in case a primary default gateway goes down. Link Aggregation Control Protocol

(LACP) is not an FHRP. LACP is an Institute of Electrical and Electronics Engineers (IEEE) protocol that is used to enable link aggregation on EtherChannel links.

## Original Exam C Question 72

Which of the following FHRPs would use the virtual MAC address 0000.5E00.0101? (Select the best answer.)

- A. only GLBP
- B. only HSRP
- C. GLBP and HSRP

**Correct Answer: D**

### Explanation

Only Virtual Router Redundancy Protocol (VRRP) would use the virtual Media Access Control (MAC) address 0000.5E00.0101. VRRP is an Internet Engineering Task Force (IETF)-standard First-Hop Redundancy Protocol (FHRP) that is supported by both Cisco and non-Cisco devices. Routers are assigned to a VRRP group, and the group functions as a single gateway for clients. AVRRP group has one master router, which is the router with the highest priority value. All other routers in the VRRP group are backup routers. The virtual MAC address for VRRP groups is in the form of 0000.5E00.01xx, where xx is a hexadecimal value identifying the VRRP group number. For example, VRRP Group 1 would be identified by the virtual MAC address 0000.5E00.0101. Gateway Load Balancing Protocol (GLBP) would not use the virtual MAC address 0000.5E00.0101. GLBP is an FHRP that also provides load balancing. GLBP enables you to configure multiple routers as a GLBP group; the routers in the group receive traffic sent to a virtual Internet Protocol (IP) address that is configured for the group. Each GLBP group contains an active virtual gateway (AVG) that is elected based on which router is configured with the highest priority value, or with the highest IP address if multiple routers are configured with the highest priority value. The other routers in the GLBP group are configured as primary or secondary active virtual forwarders (AVFs). Up to four primary AVFs can be configured in a GLBP group, and the primary AVFs can participate in forwarding traffic. Consequently, multiple routers can be used simultaneously to provide load balancing for the GLBP group. Hot Standby Router Protocol (HSRP) would not use the virtual MAC address 0000.5E00.0101. HSRP is an FHRP that is defined in Request for Comments (RFC) 2281. Similar to GLBP, HSRP can be used to provide backup router coverage if the primary gateway becomes unavailable. Multiple routers are assigned to an HSRP group, and the routers function as a single gateway. An HSRP group contains one active router and one standby router. The active router is the router with the highest priority value, and the standby router is the router with the second-highest priority value. Other routers in the HSRP group are in the listen state. If the active router fails, the standby router assumes the active router role and a new standby router is elected.

## Day 30 - TCP & UDP

Included Exam C items: Q70

## Original Exam C Question 70

Which of the following fields are found in the header of a UDP packet? (Select 4 choices.)

- A. length
- B. source port
- C. checksum
- D. sequence number
- E. window
- F. acknowledgment number
- G. destination port

**Correct Answer: A, B, C, G**

### Explanation

The source port, destination port, checksum, and length fields are found in the header of a User Datagram Protocol (UDP) packet. Each field is 16 bits, or 2 bytes, in length. UDP segment is arranged as shown below: UDP Segment 32 bits Data (variable size) The source port and destination port fields indicate the upper-layer service that will use the data. The length field

specifies the length of the UDP header and data. The checksum field is used to verify the integrity of the UDP segment to ensure that it was not modified or corrupted in transit. Because a UDP header is smaller than a TCP header, UDP requires fewer processor resources than TCP. UDP is an unreliable protocol, which means that it provides a best-effort delivery service that does not provide error correction or flow control. Therefore, the UDP header does not contain fields that are related to reliability features, such as the sequence number, acknowledgment number, and window fields. Transmission Control Protocol (TCP) is a reliable protocol. Therefore, the TCP header contains fields that provide reliability features. As a result, a TCP header is much longer than a UDP header. A TCP segment is arranged as shown below: TCP Segment 32 bits Source Port Destination Port Sequence Number Acknowledgment Number Data : : om Options (and padding) Data (variable size) The 16-bit source port and destination port fields indicate the upper-layer service that will use the data. The 32-bit sequence number field specifies the number given to the first byte of data in the segment and is used to reassemble segments into the correct order at the destination. The 32-bit acknowledgment number field specifies the sequence number of the next byte of data that the sender expects to receive. The 4-bit data offset field indicates the number of 32-bit words in the TCP header; the TCP header in the graphic above contains six 32-bit words. The 6-bit reserved field is reserved for future use and should currently be set to zero. The 6-bit flags field is used to activate control flags, such as SYN and ACK for establishing connections and FIN for terminating connections. The 16-bit window field specifies the size of the sender's receive window, which is used to buffer incoming data. The 16-bit checksum field is used to verify the integrity of the TCP segment to ensure that it was not modified or corrupted in transit. The 16-bit urgent pointer field indicates the first urgent data byte in the segment. The variable-length options field specifies other assorted TCP options; zeros are appended to pad the end of the options field so that the header ends and data begins on a 32-bit boundary.

## Days 31-33 - IPv6

Included Exam C items: Q4, Q45, Q46, Q54, Q73

### Original Exam C Question 4

Your organization is considering implementing IPv6, and you have been tasked with evaluating the feasibility of manually configuring IPv6 addresses and IPv6 routes on a staging network.

Complete the following tasks:

- Enable IPv6 routing on RouterA, RouterB, and RouterC.
- Configure FastEthernet 0/1 on RouterA with the first non-zero IPv6 address in 2001:DB8:A::/64.
- Configure FastEthernet 0/1 on RouterC with the first non-zero IPv6 address in 2001:DB8:C::/64.
- Place the RouterA-RouterB link in the first /126 subnet of 2001:DB8:B::/64. RouterB Fa0/0 should use the second address and RouterA Fa0/0 the third address.
- Place the RouterC-RouterB link in the second /126 subnet of 2001:DB8:B::/64. RouterB Fa0/1 should use the second address and RouterC Fa0/0 the third address.
- Configure RouterA and RouterC with a default route pointing to the nearest RouterB interface.
- Configure RouterB with static routes to 2001:DB8:A::/64 and 2001:DB8:C::/64.

Make only the required changes and save each router's configuration before ending the simulation.

**Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam C PDF.**

### Original Exam C Question 45

Which of the following commands can be issued to configure a Cisco router interface with the IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024 and a 96-bit subnet mask? (Select 3 choices.)

- A. ipv6 address 2001::070d::50a0:0001:0024/96
- C. ipv6 address 2001:0:0:70d:0:50a0:1:24/96
- D. ipw6 address 2001::70d:0:50a0:1:24/96
- E. ipv6 address 2001::70d::50a:1:24/96
- F. ipv6 address 2001::070d:0:50a:0001:0024/96

**Correct Answer: B, C, D**

Explanation

The following commands can be issued to configure a Cisco router interface with the Internet Protocol version 6 (IPv6) address 2001:0000:0000:070D:0000:50A0:0001:0024 and a 96-bit subnet mask:

- © ipv6 address 2001:0000:0000:070d:0000:50a0:0001:0024/96
- © ipv6 address 2001::070d:0000:50a0:0001:0024/96
- © ipv6 address 2001:0:0:70d:0:50a0:1:24/96
- © ipv6 address 2001::70d:0:50a0:1:24/96

IPv6 was developed to address the lack of available address space with IP version 4 (IPv4). An IPv6 address is a 128-bit (16-byte) address that is typically written as eight groups of four hexadecimal numbers from 0 through F. Each of four characters, including 0 through 9 A through F, is called a hexet and is separated by a colon. Leading zeros in each hexet can be dropped. A double colon can be used at the beginning, middle, or end of an IPv6 address in place of one or more contiguous hexets consisting of all zeros. However, only one double colon can be used in an IPv6 address. Each other hexet consisting of all zeroes can be replaced with a single zero. In this scenario, the IPv6 address 2001::70D:0:50A0:1:24 most efficiently expresses the expanded IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024. The command `ipv6 address 2001::070d::50a0:0001:0024/96` will not configure a Cisco router interface with the IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024 and a 96-bit subnet mask, because the address in the command is not valid. Addresses contain IPv6 a IPv6 IPv6 can only one double colon. Using two or more double colons in an address creates ambiguity because there is no way to know how many groups of zeros are represented by each double colon. The command `ipv6 address 2001::070d:0:50a:0001:0024/96` will not configure a Cisco router interface with the IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024 and a 96-bit subnet mask, because the addresses 2001::070D:0:50A:0001:0024 and 2001:0000:0000:070D:0000:50A0:0001:0024 are not equivalent. Trailing zeros in each hexet cannot be dropped. Only leading zeros in each hexet can be dropped. The command `ipv6 address 2001::70d::50a:1:24/96` will not configure a Cisco router interface with the IPv6 address 2001:0000:0000:070D:0000:50A0:0001:0024 and a 96-bit subnet mask. IPv6 addresses can contain only one double colon. In addition, trailing zeroes in each hexet cannot be dropped.

## Original Exam C Question 46

Which of the address routable?

following IPv6 prefixes are not (Select 2 choices.)

- A. FFO5::/16
- B. FC00:/8
- C. FbD00:/8
- D. FE80::/10
- E. 2000:/3
- F. FFO2::/16

**Correct Answer: D, F**

### Explanation

The Protocol version Internet 6 (IPv6) address prefixes FE80::/10 and FFO2::/16 are not routable. The IPv6 prefix FE80::/10 is used for link-local unicast addresses. IPv6 addresses in the FE80::/10 range begin with the characters FE80 through FEBF. Unicast packets are used for one-to-one communication. Link-local unicast addresses are unique only on the local segment. Therefore, link-local unicast addresses are not routable. An IPv6-capable host typically creates a link-local unicast address automatically at startup. Link-local unicast addresses are used for next-hop neighbor discovery and for environments in which no router is present to provide a routable IPv6 prefix. You can issue the `ipv6 enable` command from interface configuration mode on a Cisco router to enable the interface and IPv6 on to automatically generate a link-local unicast IPv6 address. Alternatively, you can issue the `ipv6 address` command from interface configuration mode to configure a link-local unicast IPv6 address and to enable IPv6 on the interface. The `ipv6 address` command can also be used to assign an IPv6 global unicast address to an interface, in which case IPv6 is automatically enabled and an IPv6 link-local unicast address is automatically generated. The IPv6 prefix FF02::/16 is used for link-local multicast addresses. Like link-local unicast addresses, link-local multicast addresses are not routable. There are other reserved address spaces in IPv6, such as the loopback address of ::1, which is not routable. The IPv6 prefix 2000::/3 is used for global aggregatable unicast addresses. IPv6 addresses in the 2000::/3 range begin with the characters 2000 through 3FFF. Global aggregatable unicast address prefixes are distributed by the Internet Assigned Numbers Authority (IANA) and are globally routable over the Internet. The IPv6 prefixes FC00::/8 and FDO0::/8 are used for unique local unicast addresses. These address blocks can be combined into the FC00::/7 subnet. IPv6 addresses in these ranges begin with the characters FCOO through FDFF. Like private IP version 4 (IPv4) addresses, unique local unicast addresses are not globally routable, but they are routable within an organization. All IPv6 addresses beginning with FF are multicast addresses, which are used for one-to-many communication. Ethernet frames use unicast addresses to send data to a single device, and multicast addresses are used to simultaneously send the same data to a group of devices, thereby conserving bandwidth. The following IPv6 multicast scopes are defined: e FFO1::/16 · node-local © FFO2::/16 · link-local e FFO5::/16 · site-local e FFO8::/16 · organization-local FFOE::/16 · global The FFO1::/16 prefix is used for node-local multicast addresses. These addresses are used only on the interface itself, much like a loopback address. Therefore, they are

not routable. The FF05::/16 prefix is used for site-local multicast addresses, and the FFO8::/16 prefix is used for organization-local multicast addresses. Like unique local unicast addresses, site-local multicast and organization-local multicast addresses are not globally routable, but they are routable within an organization. The FFOE::/16 prefix is used for globally routable multicast addresses. IPv6 hosts use the multicasting capabilities of the Neighbor Discovery (ND) protocol to discover the link layer addresses of neighbor hosts. The Hop Limit field is typically set to 255 in ND packets that are sent to neighbors. Routers decrement the Hop Limit value as a packet is forwarded from hop to hop. Therefore, a router that receives an ND packet with a Hop Limit value of 255 considers the source of the ND packet to be a neighbor. If a router receives an ND packet with a Hop Limit value that is less than 255, the packet is ignored, thereby protecting the router from threats that could result from the ND protocol's lack of neighbor authentication.

### Original Exam C Question 54

Your company has been assigned the 2012:0:0:99::/64 IPv6 prefix by your ISP. You have issued the following commands on your Cisco router's FastEthernet 0/0 interface:

```
ipv6 address 2012:0:0:99::/64 eui-64
no shutdown
```

Host computers should be also autoconfigured by using the modified EUI-64 format. HostA has an IPv4 address of 192.168.0.14 and a MAC address of 00-33-66-99-BB-EE.

You have configured HostA to use stateless autoconfiguration.

Which of the following IPv6 addresses will HostA use? (Select the best answer.)

- A. 2012::99:0:0233:6699:BBEE
- B. 2012::99:0:0:192.168.0.14
- C. 2012:0:0:99::COA8:000E
- D. 2012::99:0233:6699:BBEE
- E. 2012::99:0233:66FF:FE99:BBEE
- F. 2012:0:0:99:192.168.0.14

### Correct Answer: E

#### Explanation

HostA will use the Internet Protocol version 6 (IPv6) address 2012::99:0233:66FF:FE99:BBEE. The scenario indicates that hosts should be autoconfigured by using the modified extended unique identifier (EUI)-64 format. For autoconfiguration to occur, the router must send the IPv6 prefix to a host in a router advertisement message. Router advertisements, and other stateless autoconfiguration messages, are sent by using Internet Control Message Protocol version 6 (ICMPv6). An IPv6 host typically sends a router solicitation message on startup to prompt a router into sending a router advertisement, rather than waiting for the arrival of a periodic router advertisement. Once a router advertisement is received, the host will append its interface identifier, which is a modified version of its Media Access Control (MAC) address, to the received IPv6 prefix to create a globally unique, IPv6 unicast address. IPv6 was developed to address the lack of available address space with IP version 4 (IPv4). An IPv6 address is a 128-bit (16-byte) address that is typically written as eight groups of four hexadecimal characters, including numbers from 0 through 9 and letters from A through F. Each group of four characters is separated by colons. Leading zeroes in each group can be dropped. A double colon can be used at the beginning, middle, or end of an IPv6 address in place of one or more contiguous four-character groups that consist of all zeroes. However, only one double colon can be used in an IPv6 address. Therefore, the following IPv6 addresses are equivalent: e  
FE80:0000:0000:070D:0000:50A0:0001:0024 FE80::070D:0000:50A0:0001:0024 FE80:0:0:70D:0:50A0:1:24  
FE80::70D:0:50A0:1:24 An interface identifier in EUI-64 format is created by taking the first half of the host's MAC address, which is referred to as the organizationally unique identifier (OUI), adding the hexadecimal number FFFE, and then appending the last half of the host's MAC address, which is the hardware-specific portion of the MAC address. The seventh binary bit in the OUI, which is referred to as the U/L bit, is then flipped; that is, the original value of the U/L bit is inverted. A U/L bit value of 0 in the OUI indicates that the MAC address is universal. Universal addresses are intended to be globally unique and are the addresses burned in by the manufacturer. By contrast, a U/L bit value of 1 indicates a locally administered MAC address, such as the address created by a virtual interface or manually configured by an administrator. The U/L value is inverted when an EUI-64 interface ID is created in order to facilitate simple local scope identifiers for manual administration. It should be noted that there is no correlation between the U/L bit value in an EUI-64 interface ID and the scope of the IPv6 address. The U/L bit value in an EUI-64 interface ID was intended to provide the ability for future technology to identify interface IDs with a local scope. In this scenario, the MAC address 00-33-66-99-BB-EE has an OUI of 00-33-66 and a network interface card (NIC) identifier of 99-BB-EE. To create an EUI-64-compliant interface ID, you should first append the hexadecimal value FFFE to the OUI and then append the NIC identifier: 0033:66FF:FE99:BBEE. Finally, you should invert the value of the seventh bit of the OUI to represent the scope of the EUI-64 interface ID. The hexadecimal value 00 in the first 8 bits of the OUI can be

represented in binary as 0000 0000 and indicates that the seventh bit represents the U/L bit. In this scenario, the seventh bit has a value of 0, which indicates that the MAC address from which the EUI-64 address is derived is a universal address. The value of the bit the value from the first Inverting U/L changes 0 to 1; therefore, eight bits of the OUI become 0000 0010. These bits can be represented in hexadecimal as 02, and they result in an EUI-64 interface ID of 0233:66FF:FE99:BBEE. The EUI-64 interface ID is then combined with the IPv6 prefix to create an IPv6 address. Appending the IPv6 prefix 2012::99 to this interface identifier creates the global unicast IPv6 address 2012::99:0233:66FF:FE99:BBEE. This address is considered globally unique because the IPv6 prefix is in the 2000::/3 range. IPv6 global unicast addresses always begin with a 2 or a 3 because the first three bits of an IPv6 global unicast address are always 001. HostA will not use the IPv6 address 2012::99:0233:6699:BBEE. The address 2012::99:0233:6699:BBEE expands to 2012:0:0:0:99:0233:6699:BBEE, which does not use the correct prefix value of 2012:0:0:99::/64. HostA will not use the IPv6 address 2012::99:0:0033:6699:BBEE. Although the interface identifier is composed by using the MAC address, it does not follow the EUI-64 format. HostA will not use the IPv6 addresses 2012::99:0:0:192.168.0.14 or 2012:0:0:99::COA8:000E. An IPv4-compatible IPv6 address can be created by using zeroes for the first 96 bits of the address and by using the IPv4 address for the last 32 bits of the address. The IPv4 address can be written with or without leading zeroes and can be written in binary or hexadecimal format. Therefore, the following notations would be acceptable for HostA if it were to use an IPv4-compatible IPv6 address: © 0:0:0:0:0:192.168.0.14 © ::192.168.0.14 © 0:0:0:0:0:COA8:000E e :COA8:000E When you convert the IPv4 address 192.168.0.14 from decimal to hexadecimal, 192 converts to CO, 168 converts to A8, 0 converts to 00, and 14 converts to OE. HostA will not use the IPv6 address 2012:0:0:99:192.168.0.14, because it does not contain enough bits to create a valid 128-bit IPv6 address. The prefix 2012:0:0:99::/64 contains 64 bits for the prefix, and 64 bits are needed for the interface identifier. IPv4 addresses contain only 32 bits, so appending 192.168.0.14 directly to the prefix creates an address with only 96 bits.

### Original Exam C Question 73

issue the command You following on a Cisco router in global configuration mode:

```
ipv6 route 2001:db8:a::/32 fastethernet 0/1
```

Which of the following have you created? (Select the best answer.)

- A. a directly attached static route
- B. a fully specified static route
- C. a floating static route
- D. a recursive static route

**Correct Answer: A**

Explanation

have created attached Protocol version You a directly Internet 6 (IPv6) static route if you issue the fastethernet command Cisco in `ipv6 route 2001:db8:a::/32 0/1` on a router global configuration mode. A directly attached static route specifies the destination IPv6 network and the outbound interface. When connected static is the that a directly route configured, router assumes any packet that matches the destination network is reachable through the specified outbound interface. Therefore, the packet's full destination address is used as the IPv6 next-hop address. There are three other types of IPv6 static routes: fully specified static routes, recursive static routes, and floating static routes. A fully specified static route is an IPv6 static route in which the destination network, outbound interface, and next-hop IPv6 address are all configured directly. For example, if you were to issue the `ipv6 route 2001:db8:a::/32 fastethernet 0/1 2001:db8:b::1` command, the IPv6 prefix of 2001:db8:a::/32 represents the destination network. The router's FastEthernet 0/1 interface is the outbound interface. The IPv6 address of 2001:db8:b::1 is the next-hop IPv6 address. Fully specified static routes are most often used when the outbound interface is multiaccess and could therefore be configured with multiple next-hop addresses. The next-hop address that is specified in the command must be directly connected to the outbound interface. A recursive static route specifies the destination IPv6 network and the IPv6 next-hop address only. For example, the `ipv6 route 2001:db8:a::/32 2001:db8:a::1` command configures the router to resolve all IPv6 addresses in the 2001:db8:a::/32 prefix through the next hop that has been assigned the IPv6 address of 2001:db8:a::1. The router assumes the outbound interface to be the interface to which the next hop is either directly or indirectly connected. In other words, the next-hop IPv6 address must be resolvable through the outbound interface. static be of any of the other three of IPv6 static A floating route can comprised types routes. A floating static route is typically used as a backup route when the primary route goes down. Floating static routes are typically configured with an administrative distance (AD) value that is numerically higher than the AD of the primary route. For example, the `ipv6 route 2001:db8:a::/32 2001:db8:a::1 5` command will install a static route with an AD of 5. If a static route with the default AD of 1 were to go down, the floating static route could be used as a backup route to the destination network.

## Days 34-35 - Access Control Lists (ACLs)

Included Exam C items: Q5, Q7, Q49, Q61

### Original Exam C Question 5

Drag the commands to the appropriate descriptions. Commands can be used more than once. Not all commands will be used.

Commands: access-class; ipv6 access-class; ip access-group; ipv6 traffic-filter.

Descriptions: applies an IPv4 ACL to the console; applies an IPv6 ACL to the console; applies an ACL to an IPv4 interface; applies an ACL to an IPv6 interface.

**Correct Answer: access-class -> applies an IPv4 ACL to the console/vty/AUX line**

**ipv6 access-class -> applies an IPv6 ACL to the console/vty/AUX line**

**ip access-group -> applies an ACL to an IPv4 interface**

**ipv6 traffic-filter -> applies an ACL to an IPv6 interface**

#### Explanation

The ip access-group command can be used to apply an access control list (ACL) to an Internet Protocol version interface. Each interface with inbound 4 (IPv4) You can configure on a router one ACL and one outbound ACL. Incoming traffic is checked against inbound ACLs. To configure an inbound ACL for an interface, you should issue the ip access-group list-number in command in interface configuration mode. Outgoing traffic is checked against outbound ACLs. To configure an outbound ACL for an interface, you should issue the ip access-group list-number out command in interface configuration mode. The ipv6 traffic-filter command can be used in interface configuration mode to apply an ACL to an IP version 6 (IPv6) network interface. The syntax for the ipv6 traffic-filter command is ipv6 traffic-filter ipv6-acl [in | out], where the in keyword configures the ACL to apply to inbound traffic and the out keyword configures the ACL to apply to outbound traffic. The access-class command can be used to apply an IPv4 ACL to a virtual terminal (vty) line, the auxiliary (AUX) line, or the console line. The syntax for the access-class command is access-class acl-name-or-number [in | out], where the in keyword configures the ACL to apply to inbound traffic and the out keyword configures the ACL to apply to outbound traffic. Likewise, the ipv6 access-class command can be used to apply an IPv6 ACL to a vty line, the AUX line, or the console line. The syntax for the ipv6 access-class command is ipv6 access-class ipv6-acl [in | out].

### Original Exam C Question 7

Issue the commands You following on a Cisco router named RouterA:

```
RouterA(config)#ip access-list extended boson
```

```
RouterA(config-ext-nacl)#deny tcp any any neq 23
```

```
RouterA(config-ext-nacl)#permit tcp any any eq 20 21
```

```
RouterA(config-ext-nacl)#permit tcp any any range 22 80
```

```
RouterA(config-ext-nacl)#permit tcp any any gt 442
```

Which of the following statements about the ACL is true? (Select the best answer.)

- A. Traffic that matches TCP destination port 23 will be permitted. Traffic that matches TCP destination ports in the range from 22 through 80 will be permitted.
- B. Traffic that matches TCP destination port 20 or TCP destination port 21 will be permitted.
- C. Traffic that matches TCP destination port 20 or TCP destination port 21 will be permitted.
- D. Traffic that matches TCP destination port 443 will be permitted.

**Correct Answer: A**

#### Explanation

Traffic that matches Transmission Control Protocol (TCP) destination port 23 will be permitted by the named extended access control list (ACL) that is configured in this scenario. To create a named extended ACL, you should issue the ip access-list extended acl-name command. Issuing this command will place the in extended-named-ACL mode. When router A packet matches an ACL rule, the ACL will not attempt to match the traffic to any subsequent rules that have been configured in the ACL. Extended ACL entries can be created by using the following basic syntax: [sequence-number] {deny | permit} protocol source source-wildcard [operator port [port]] destination destination-wildcard [operator port [port]] The optional sequence-number value enables an administrator to specify a sequence number to insert a rule into a specific order in an existing ACL. If a value is not specified for a sequence number, the ACL will automatically add a sequence number in an interval of 10. In addition, port numbers that map to ACL keywords will be converted to those keywords, as shown in the following output:

RouterA#show ip access-lists Extended IP access list boson 10 deny tcp any any neq telnet 20 permit tcp any any eq ftp-data ftp 30 permit tcp any any range 22 www 40 permit tcp any any gt 442 In the output above, the first rule has been assigned a sequence number of 10. The second rule's number is an increment of 10, and add rule the of the sequence so on. To a to beginning existing ACL above, an administrator could issue the rule with a sequence-number value of 5. In addition, ACL keywords that map to specific port numbers have replaced those port numbers in the output. Port 23 has an equivalent keyword of telnet. Port 80 has an equivalent keyword of www. The File Transfer Protocol (FTP) ports 20 and 21 have equivalent keywords of ftp-data and ftp, respectively. In this scenario, the deny tcp any any neq 23 command creates an extended ACL rule that denies a TCP packet that does not have a destination port value of 23, regardless of the source address of that packet or the destination address of that packet. The tcp protocol keyword configures the rule to match TCP packets. The source wildcard value of any configures the rule to match the TCP packet regardless of the source address. The destination wildcard value of any configures the rule to match the TCP packet regardless of the destination address. The neq operator matches traffic that is not equal to the port number that is specified after the operator, which is port 23 in this scenario. Because TCP packets that do not have a destination port value of 23 will always match the first rule in the ACL in this scenario, any traffic that is not destined to TCP port 23 will be denied. Traffic that matches TCP destination port 20 or TCP destination port 21 will not be permitted in this scenario, because the permit tcp any any eq 20 21 rule follows the deny tcp any any neq 23 rule in the ACL. If the deny tcp any any neq 23 rule were not configured, the permit tcp any any eq 20 21 rule would match TCP packets from any source to any destination as long as the TCP destination port is equal to either port 20 or port 21. The eq operator matches packets that have destination ports that are equal to the port number that is specified after the operator. In addition, up to 10 port numbers separated by spaces can be specified after the eq or neq operators. Traffic that matches TCP destination ports in the range from 22 through 80 will not be permitted in this scenario, because the permit tcp any any range 22 80 rule follows the deny tcp any any neq 23 rule in the ACL. If the deny tcp any any neq 23 rule were not configured, the permit tcp any any range 22 80 rule would match TCP packets from any source to any destination as long as the TCP destination port is equal to a port number in the range from port 22 through port 80. The range operator matches packets that have destination ports that are equal to the port number specified as the start of the range of ports, the port number specified as the end of the range of ports, and any port number that falls in between those two port numbers. The range operator is the only operator that requires two values. Traffic that matches TCP destination port 443 will not be permitted in this scenario, because the permit tcp any any gt 442 rule follows the deny tcp any any neq 23 rule in the ACL. If the deny tcp any any neq 23 rule were not configured, the permit tcp any any gt 442 rule would match TCP packets from any source to any destination as long as the TCP destination is greater than port number 442. The rule would not match packets that have a destination port of 442 or lower. The gt operator matches packets that have destination ports that are greater than the port number specified. Conversely, the lt operator matches packets that have destination ports that are less than the port number specified. Only one port number value can be specified after the gt and lt operators. In addition, you cannot combine the gt and lt operators to specify a range.

## Original Exam C Question 49

· \_ - · f 27.246.

96/50 So/o Fa0/0 Operations 10.10.10.0/24 Email Server Web Server 10.10.10.10/24 10.10.10.11/24 Your company has added an email server and a web server to the existing network infrastructure. Company policy stipulates that only inbound SMTP traffic be allowed to pass to Email Server and that only inbound web traffic be allowed to pass to Web Server. All other traffic must be prevented from reaching the subnet.

Which access lists should be applied to the S0/0 interface of Router1? (Select 2 choices.)

- A. access-list 101 permit udp any 10.10.10.10 0.0.0.0 eq 25
- access-list 101 permit tcp any 10.10.10.11 0.0.0.0 eq 80
- C. access-list 101 permit tcp any 10.10.10.10 0.0.0.255 eq 25
- D. access-list 101 permit tcp any 10.10.10.11 0.0.0.255 eq 80
- E. access-list 101 permit udp any 10.10.10.11 0.0.0.0 eq 80
- access-list 101 permit tcp any 10.10.10.10 0.0.0.0 eq 25

**Correct Answer: B, F**

### Explanation

You should apply the following access control lists (ACLs) to the S0/0 interface of Router1: access-list 101 permit tcp any 10.10.10.11 0.0.0.0 eq 80 access-list 101 permit tcp any 10.10.10.10 0.0.0.0 eq 25 An individual ACL is applied to a specific interface and in one direction. In this case, two ACLs should be applied to the S0/0 interface of Router1, which is the interface to which inbound traffic must first travel from the Internet to the Operations network. Whereas standard ACLs can be used to filter only on source Internet Protocol (IP) addresses, extended ACLs can be used to filter on source and destination IP addresses, protocols, and ports. In this scenario, you are required to filter based on destination and port, so you must use an extended ACL. Extended ACLs require the use of source and destination IP addresses. These IP addresses can cover single

hosts or large ranges of hosts depending on the wildcard masks specified. On Cisco platforms running IOS 12.0.1 or later, extended ACLs can have access list numbers from 100 through 199 and from 2000 through 2699; the ACL number range 2000-2699 is referred to as the expanded range. ACLs are constructed by using the following syntax: Standard: access-list access-list-number [permit | deny] [host | source source-wildcard | any] Extended: access-list access-list-number [dynamic dynamic-name [timeout minutes] {deny | permit} protocol source source-wildcard [operator [port]] destination destination-wildcard [operator [port]] [established] [precedence precedence] [tos tos] [log | log-input] [time-range time-range-name] Named: ip access-list {extended | standard} name After you issue the ip access-list standard name command, the device will be placed in standard ACL configuration mode, where you can issue one or more statements to configure the named ACL. Similarly, issuing the ip access-list extended name command will place the device into extended ACL configuration mode. Email servers use Simple Mail Transfer Protocol (SMTP) to transfer messages over the Internet. Because SMTP operates at port 25 of the Transmission Control Protocol/IP (TCP/IP) stack by default, the access-list 101 permit tcp any 10.10.10.10 0.0.0.0 eq 25 command should be applied to the S0/0 interface of to allow inbound connections to the email server. Named ACLs are different from standard or extended ACLs because they are identified by a string of characters instead of a number. In addition, a named ACL can be configured as either a standard ACL or an extended ACL. Although you could configure a named ACL to satisfy the requirements in this scenario, none of the choices includes a named ACL configuration. All access lists contain an implicit deny rule, which denies traffic that has not been explicitly allowed. Therefore, you do not need to issue any other commands to deny Internet communications. Web servers typically operate at port 80 of the TCP/IP stack. Thus the access-list 101 permit tcp any 10.10.10.11 0.0.0.0 eq 80 command should be applied to the S0/0 interface of Router1 to allow web traffic to access the web server. The access-list 101 permit tcp any 10.10.10.10 0.0.0.255 eq 25 command would allow all hosts on the 10.10.10.0 network to receive SMTP traffic because of the 0.0.0.255 wildcard mask. To ensure that only the email server is a valid destination, you should use the wildcard mask 0.0.0.0. Similarly, the access-list 101 permit tcp any 10.10.10.11 0.0.0.255 eq 80 command would allow all hosts on the 10.10.10.0 network to receive web traffic. To ensure that web traffic is allowed to reach only the web server, you should use the 0.0.0.0 wildcard mask. The access-list 101 permit udp any 10.10.10.10 0.0.0.0 eq 25 and access-list 101 permit udp any 10.10.10.11 0.0.0.0 eq 80 commands would allow User Datagram Protocol (UDP) traffic, not TCP traffic. SMTP and Hypertext Transfer Protocol (HTTP) use TCP, not UDP, for host communications. UDP is commonly used by applications that are connectionless by design, such as Trivial File Transfer Protocol (TFTP). Valid options for the protocol constraint of an ACL include ip, tcp, udp, icmp, and others.

## Original Exam C Question 61

Internet You administer the network shown in the exhibit. You have created an inspection rule named boson and applied the inspection rule to the untrusted interface in the outbound direction.

You want to protect the internal network from external threats while allowing legitimate IP traffic from the internal network to communicate with the external network.

Which of the following command sets should you issue? (Select the best answer.)

```
RouterA(config)#access-list 101 deny ip 1.2.3.4 0.0.0.3 any
RouterA(config)#access-list 101 deny ip host 255.255.255.255 any
RouterA(config)#access-list 101 permit ip any any
RouterA(config)#interface se/e
RouterA(config-if)#ip access-group 101 in
RouterA(config)#access-list 102 deny ip 192.168.10.0 0.0.0.255 any log
RouterA(config)#access-list 102 permit icmp any host 1.2.3.5 unreachable
RouterA(config)#access-list 102 deny ip any any
RouterA(config)#interface s0/0
RouterA(config-if)#ip access-group 102 in
RouterA(config)#access-list 102 deny ip 192.168.10.0 0.0.0.255 any log
RouterA(config)#access-list 102 permit icmp any host 1.2.3.5 unreachable
RouterA(config)#access-list 102 deny ip any any
RouterA(config)#interface fa1/0
RouterA(config-if)#ip access-group 102 in
RouterA(config)#access-list 101 deny ip 1.2.3.4 6.0.0.3 any
RouterA(config)#access-list 101 deny ip host 255.255.255.255 any
RouterA(config)#access-list 101 permit ip any any
RouterA(config)#interface fa1/0
RouterA(config-if)#ip access-group 101 out
```

**Correct Answer: B**

Explanation

Of the choices provided, you should issue the following set of commands to protect the internal network from external threats while allowing legitimate Internet Protocol (IP) traffic from the internal network to communicate with the external network:

```
RouterA(config)#access-list 102 deny ip 192.168.10.0 0.0.0.255 any log RouterA(config)#access-list 102 permit icmp any host 1.2.3.5 unreachable RouterA(config)#access-list 102 deny ip any any RouterA(config)#interface se/e RouterA(config-if)#ip access-group 102 in
```

The commands above configure an extended access control list (ACL) named 102 that denies and logs attempts to spoof IP traffic from the 192.168.10.0/24 network, permits Internet Control Message Protocol (ICMP) unreachable messages destined for the IP address 1.2.3.5, and denies all other inbound IP traffic on the S0/0 interface of RouterA, which is the untrusted interface. Because 192.168.10.0/24 is the network address of the inside network, packets should never arrive on the external interface from an IP address in this range. Denying this traffic prevents IP address spoofing. In addition, the threat of ICMP-based attacks on the internal network is reduced because only ICMP unreachable messages destined to 1.2.3.5 are permitted. Typically, you should deny all unnecessary inbound traffic on an untrusted interface to protect the internal network from external threats. You should not issue the following set of commands to protect the internal network from external threats:

```
RouterA(config)#access-list 102 deny ip 192.168.10.0 0.0.0.255 any log RouterA(config)#access-list 102 permit icmp any host 1.2.3.5 unreachable RouterA(config)#access-list 102 deny ip any any RouterA(config)#interface fa1/e RouterA(config-if)#ip access-group 102 in
```

The commands above apply ACL 102 in the inbound direction on interface Fa1/0, which is the trusted interface on RouterA. Because ACL 102 denies traffic from the trusted network 192.168.10.0/24, this configuration does not allow legitimate IP traffic from the internal network to communicate with the external network. If you were to issue the ip access-group 102 out command on interface Fa1/0 instead of the ip access-group 102 in command, this ACL would protect the internal network. You should not issue the following set of commands to protect the internal network from external threats:

```
RouterA(config)#access-list 101 deny ip 1.2.3.4 0.0.0.3 any RouterA(config)#access-list 101 deny ip host 255.255.255.255 any RouterA(config)#access-list 101 permit ip any any RouterA(config)#interface s0/0 RouterA(config-if)#ip access-group 101 in
```

The commands above configure an inbound extended ACL named 101 on interface S0/0. ACL 101 denies IP traffic from the IP addresses 1.2.3.5 and 1.2.3.6 to any destination. In addition, ACL 101 denies traffic from the broadcast address 255.255.255.255 to any destination. ACL 101 permits all other IP traffic. The IP addresses 1.2.3.5 and 1.2.3.6 are the endpoints of the point-to-point connection from RouterA to the Internet. Because these addresses are denied, all traffic originating from the service provider's router is discarded. However, any other unicast IP traffic is permitted to pass through the S0/0 interface. This configuration exposes the internal network to threats from the external network instead of protecting it from threats. Finally, you should not issue the following set of commands to protect the internal network from external threats:

```
RouterA(config)#access-list 101 deny ip 1.2.3.4 any RouterA(config)#access-list 101 deny ip host 255.255.255.255 any RouterA(config)#access-list 101 permit ip any any RouterA(config)#interface fa1/0 RouterA(config-if)#ip access-group 101 out
```

The commands above apply ACL 101 in the outbound direction on the secure interface, Fa1/0. Although ACL 101 does not hinder the internal network from sending legitimate IP traffic to the external network, the ACL does not protect the internal network from external threats.

## Day 36 - CDP & LLDP

Included Exam C items: Q26, Q52

### Original Exam C Question 26

Which of the following statements are not true about CDP? (Select 2 choices.)

A. It can be configured at the interface level to receive CDP packets but not transmit them.

B. B. It can be disabled at the interface level. It can be disabled globally. C. C. It can be enabled if LLDP is already enabled. D. D. It can be configured at the interface level to transmit CDP packets but not receive them.

**Correct Answer: A, E**

#### Explanation

Cisco Discovery Protocol (CDP) cannot be configured at the interface level to only receive CDP packets, nor can it be configured at the interface level to only transmit CDP packets. CDP is a Cisco-proprietary discovery protocol that operates at the data link layer of the Open Systems Interconnection (OSI) model. CDP can be disabled either globally or per interface. You should issue the no cdp run command from global configuration mode to disable CDP globally on a device. To re-enable CDP globally on a router or switch, you should issue the cdp run command. You should issue the enable command from interface configuration mode to enable CDP on a single interface. To re-enable CDP on an interface, you should issue the cdp enable command from interface configuration mode. By contrast, Link Layer Discovery Protocol (LLDP) is an OSI Layer 2 open-

standard discovery protocol that can be configured to control the sending and receiving of LLDP packets for a specific interface. Like CDP, LLDP can be disabled globally or at the interface level. You can disable LLDP globally by issuing the `no lldp run` command. You can re-enable LLDP by issuing the `lldp run` command. On an interface, you can disable LLDP by issuing the `no lldp enable` command. You can re-enable LLDP on the interface by issuing the `lldp enable` command. By default, all interfaces are configured to transmit and receive LLDP packets when LLDP is globally enabled. However, you can issue the `lldp transmit` and `lldp receive` commands from interface configuration mode to control the sending and receiving of LLDP packets for a specific interface. For example, the `no lldp transmit` command configures an interface not to send LLDP packets, whereas the `lldp receive` command configures an interface that was previously configured not to receive LLDP packets to resume listening for LLDP packets. Issuing the `no lldp transmit` and `no lldp receive` commands from interface configuration mode will effectively disable LLDP on an interface, but it will not globally disable LLDP. Both CDP and LLDP can be configured to run on a Cisco device without any conflicts.

## Original Exam C Question 52

Which of the following commands should you issue to globally disable LLDP? (Select the best answer.)

- A. `nolldp receive`
- B. `B. lldp holdtime 0`
- D. `nolldp transmit`

**Correct Answer: C**

### Explanation

You should issue the `no lldp run` command from global configuration mode to disable Link Layer Discovery Protocol (LLDP). LLDP is an Open Systems Interconnection (OSI) Layer 2 open- standard discovery protocol that is used to facilitate interoperability between Cisco devices and non- Cisco devices. Attributes that can be learned from neighboring devices contain Type, Length, Value (TLV) information including port description, system description, and management address. LLDP is disabled by default on Cisco switches. By default, all interfaces are configured to transmit and receive LLDP packets when LLDP is globally enabled. However, you can issue the `lldp transmit` and `lldp receive` commands from interface configuration mode to control the sending and receiving of LLDP packets for a specific interface. For example, the `no lldp transmit` command configures an interface not to send LLDP packets, whereas the `lldp receive` command configures an interface that was previously configured not to receive LLDP packets to resume listening for LLDP packets. Issuing the `no lldp transmit` and `no lldp receive` commands from interface configuration mode will effectively disable LLDP on an interface, but it will not globally disable LLDP. By default, a Cisco switch will send LLDP advertisements every 30 seconds when LLDP is enabled on an interface. These advertisements are used by neighboring devices to update the LLDP information learned about each neighbor. They are also used as keepalive messages to ensure that a discovered neighbor continues to be available on the network. You can issue the `lldp timer rate` command from global configuration mode to configure the frequency at which LLDP advertisements are sent by a switch. The default rate value is 30 seconds; however, the rate can be configured to any integer value from 5 through 65534 seconds. A Cisco switch will retain LLDP information for 120 seconds when LLDP is enabled on an interface. This time interval is known as the LLDP holdtime. You can issue the `lldp holdtime seconds` command from global configuration mode to configure the LLDP holdtime to any integer value from 0 through 65535 seconds. Whenever a new LLDP advertisement is received, the hold timer is reset and the LLDP information is considered current. When the hold timer expires for a particular neighbor, the LLDP information regarding that neighbor is considered stale and is discarded. Issuing the `lldp holdtime 0` command configures the hold timer to expire immediately; however, it will not globally disable LLDP.

## Day 37 - NTP

Included Exam C items: Q71

## Original Exam C Question 71

Which of the following commands should you issue from global configuration mode to configure a Cisco router to operate as an NTP server? (Select the best answer.)

- A. `ntp server`
- B. `ntp broadcastclient`
- D.

ntp peer  
O ntp authenticate

**Correct Answer: C**

**Explanation**

You should issue the ntp master command from global configuration mode to configure a Cisco router to operate as a Network Time Protocol (NTP) server. NTP is used to synchronize the time on network devices. The syntax of the ntp master command is ntp master [stratum], where stratum is an NTP stratum value from 1 through 15. By default, an NTP server is configured to use a stratum value of 8. NTP servers not only synchronize time with NTP clients but also synchronize time with each other. Devices with higher stratum numbers receive time from devices with lower stratum numbers. For example, a stratum 2 device typically receives its time from a stratum 1 device, a stratum 3 device typically receives its time from a stratum 2 device, and so on. You should not issue the ntp server command from global configuration mode to configure a Cisco router to operate as an NTP server. Issuing the ntp server command from global configuration mode configures a Cisco router to operate as an NTP static client. An NTP static client receives its time from an NTP server. The syntax of the ntp server command is ntp server ip-address, where ip-address is the Internet Protocol (IP) address of the NTP server that the client will use to receive its time. You should not issue the ntp broadcast client command from global configuration mode to configure a Cisco router to operate as an NTP server. Issuing the ntp broadcast client command from interface configuration mode configures a Cisco router to operate as an NTP broadcast client. An NTP broadcast client listens on the configured interface for NTP broadcasts from an NTP server, which the client its time. The difference between broadcast client and NTP uses to adjust a static client is that a broadcast client can receive its time from any NTP server. By contrast, a static client receives its time from the NTP server specified in the ntp server command. You should not issue the ntp authenticate command from global configuration mode to configure a Cisco router to operate as an NTP server. Issuing the ntp authenticate command from global configuration mode enables NTP authentication on a Cisco router. Authentication can be used with NTP to provide source verification for NTP synchronization. To configure the router with the NTP Message Digest 5 (MD5) authentication key, you should issue the ntp authentication-key key- number md5 key [encryption-type] command. You should not issue the ntp peer command from global configuration mode to configure a Cisco router to operate as an NTP server. Issuing the ntp peer command from global configuration mode configures a Cisco router to operate in NTP symmetric active mode. A router in symmetric active mode attempts to mutually synchronize with another NTP host; the host might synchronize the peer, or it might be synchronized by the peer. The syntax of the ntp peer command is ntp peer ip-address, where ip-address is the IP address of the NTP host.

## Day 38 - DNS

Included Exam C items: Q6

### Original Exam C Question 6

Which of the following commands enables a Cisco router to send DNS queries? (Select the best answer.)

- A. ip dns server
- B. ip host
- C. ip name-server
- D. ip domain lookup

**Correct Answer: D**

**Explanation**

Issuing the ip domain lookup command in global configuration mode enables a Cisco router to send Domain Name System (DNS) queries. The ip domain lookup command configures the router to use DNS to translate host names into Internet Protocol (IP) addresses. DNS is a mechanism that translates the host name of a network device into the corresponding IP address. DNS clients send lookup queries to a DNS server on User Datagram Protocol (UDP) port 53 unless the query is larger than 512 bytes. If the query is larger than 512 bytes, the DNS traffic is sent to Transmission Control Protocol (TCP) port 53. The response to each query is cached by the DNS client to facilitate faster lookups. To disable DNS lookup queries, you should issue the no ip domain lookup command on the DNS client. The ip name-server command by itself does not enable a Cisco router to send DNS queries. The ip name-server server-address1 [server-address2...server-address6] command specifies a list of DNS servers that a Cisco router can connect to in order to perform DNS lookups. However, issuing the ip name-server

command without DNS queries enabled will result in translation failures. You should not issue the `ip dns server` command to enable a Cisco router to send DNS queries; this command is used to configure a Cisco router as a DNS server. The `ip dns server` command enables the router to respond to any DNS queries that it receives from other devices on the network. You should not issue the `ip host` command to enable a Cisco router to send DNS queries; this command is used to configure a local host name-to-address mapping. The `ip host host-name ip-address` command associates different 1 [ip-address2...ip-address8] configures router to up to eight IP addresses with a single host name. By default, a router configured as a DNS server will use only its local host name-to-address mappings to resolve DNS queries. However, if the router is also configured with both the `ip domain lookup` command and the `ip name-server` command, it will forward DNS queries that it cannot resolve locally to one of its configured name servers.

## Day 39 - DHCP

Included Exam C items: Q1, Q15

### Original Exam C Question 1

You want to configure RouterB as a DHCP server for hosts that are connected to the FastEthernet 0/1 interface of RouterA. The IP address of the FastEthernet 0/1 interface of RouterA should be reserved; the remainder of the subnet should be available for dynamic IP address allocation.

Your supervisor has asked you to complete the following tasks:

- Configure a pool named boson on RouterB to assign IP addresses for the hosts on the network connected to the FastEthernet 0/1 interface of RouterA.
- Ensure that the hosts are configured with the correct default gateway.
- Configure RouterA so that DHCP requests are forwarded to RouterB.

**Source result: NetSim lab task - no separate answer/explanation is included in the uploaded Exam C PDF.**

### Original Exam C Question 15

You want to configure your Cisco router to provide IP addresses to the computers on your network. The IP addresses should be assigned from the 192.168.1.0/26 address range.

Which of the following commands should you issue? (Select 2 choices.)

- A. `ip address dhcp`
- C. `network 192.168.1.0 255.255.255.192`
- D. `network 192.168.1.0 0.0.0.63`
- E. `host 192.168.1.0 255.255.255.192`
- F. `dhcp pool 1`

**Correct Answer: B, C**

Explanation

To configure your Cisco router to provide Internet Protocol (IP) addresses to the computers on your network over Dynamic Host Configuration Protocol (DHCP), you should issue the `ip dhcp pool 1` command from global configuration mode and then issue the `network 192.168.1.0 255.255.255.192` command from DHCP pool configuration mode. The router will then provide IP addresses to hosts connected to the router interface that belongs to that subnet. The syntax of the `ip dhcp pool` command is `ip dhcp pool name`, where name is the name of your DHCP pool. The syntax of the `network` command is `network address [mask | /prefix]`, where address is the network address, mask is the subnet mask, and prefix is the prefix length in Classless Inter-Domain Routing (CIDR) notation. You should not issue the `dhcp pool 1` command, because it contains invalid syntax. To configure a Cisco router to become a DHCP server, you must issue the `ip dhcp pool name` command. You should not issue the `network 192.168.1.0 0.0.0.63` command, because the subnet mask is incorrectly specified as a wildcard mask. To configure your DHCP pool with addresses from the 192.168.1.0/26 range, you should issue the `network 192.168.1.0 255.255.255.192` command. Alternatively, you can configure the DHCP pool using CIDR notation by issuing the `network 192.168.1.0 /26` command. You should not issue the `host 192.168.1.0 255.255.255.192` command. The `host` command, when issued from DHCP pool configuration mode, is used to configure an IP address for a manual binding. A manual binding enables a device to always receive the same IP address from DHCP by associating a static IP address with the device's Media Access Control

(MAC) address. To configure a manual binding, you should issue the `host address [mask | /prefix]` command from DHCP pool configuration mode, where address is the address of the device, mask is the subnet mask, and prefix is the prefix length in CIDR notation. Then you should issue the `client-identifier MAC` command, where MAC is the client's MAC address in dotted hexadecimal notation. For example, to create a manual binding so that the computer with MAC address 0000.0c12.3456 always receives the IP address 192.168.1.20/26, you should issue the following commands: `host 192.168.1.20 /26 client-identifier 0000.0c12.3456` Alternatively, you can issue the `host 192.168.1.20 255.255.255.192` command to specify the IP address for the static mapping. You cannot use the same DHCP pool for manual bindings and for dynamic IP address allocation. You should not issue the `ip address dhcp` command. The `ip address dhcp` command configures an interface to become a DHCP client so that it can receive IP configuration information from a DHCP server. A DHCP client can receive an IP address, a subnet mask, a domain name, a Domain Name System (DNS) server, and more from a DHCP server.

## Day 40 - SNMP

Included Exam C items: Q33

### Original Exam C Question 33

Drag each term to its appropriate definition. All terms should be used.

Terms: Agent, ASN.1, MIB, NMS, Probe, RMON, RMON2, SNMP.

**Correct Answer: SNMP -> a standard protocol used to convey management information**

**MIB -> a database that contains information about a managed device**

**ASN.1 -> a set of rules for storing data in a machine-independent format**

**Agent -> a software process that runs on a managed device and gathers management information**

**Probe -> a standalone device that can monitor a network segment**

**NMS -> a central system that proactively gathers management information from managed devices**

**RMON -> an extension to the MIB that adds statistical and historical Physical/Data Link layer information**

**RMON2 -> an extension to the MIB that adds statistical and historical Network/Application layer information**

Explanation

Simple Network Management Protocol (SNMP) is a protocol that a Network Management System (NMS) can use to communicate with an agent. An agent is a software process that runs on a managed device and information. also standalone gathers device-specific management An agent can run on a device called a probe, which monitors a network segment. The Management Information Base (MIB) is a database that contains information about a managed device. A managed device can locally store information in its MIB. When the device is queried by an NMS, the device can reference its MIB and return the required value to the NMS. The MIB stores information about a managed device in a standard format specified by Abstract Syntax Notation One (ASN.1). ASN.1 is a set of rules for storing data in a machine-independent format. ASN.1 was developed by the International Telecommunication Union · Telecommunication Standardization Sector (ITU-T) in order to define a standard language for storing data. Because the information is stored in a machine-independent format, the information is guaranteed to be accessible by any SNMP-compliant management station. Remote Network Monitoring (RMON) is an extension of the MIB data structure that adds statistical and historical Physical layer and Data Link layer information. RMON enables a managed device to locally store statistical and historical data. Because an RMON-capable device can store its own statistical data, the number of queries by an NMS is reduced. Similarly, RMON2 is an extension of the MIB data structure that adds statistical and historical Network layer and Application layer information.

## Day 41 - Syslog

Included Exam C items: Q24

## Original Exam C Question 24

You receive the following output on the console of RouterA:

1 14:20:07.059: %DUAL-5-NBRCHANGE: IP-EIGRP(00) 1: Neighbor 192.168.1.2 (FastEthernet0/0) is down: interface down  
Which of the following statements is true? (Select the best answer.)

- A. The service timestamps debug datetime command has been issued on RouterA.
- B. B. The logging history size 1 command has been issued on RouterA.
- C. The default logging facility code is being displayed in the output.
- D. A notification-level severity code is being displayed in the output.

### Correct Answer: D

#### Explanation

A notification-level severity code is being displayed in the output. Notification-level events include route flaps, neighbor adjacencies, interface protocol up/down transitions, and system restart messages. Cisco debug messages and log messages are divided into the following severity levels: © 0-emergencies e 1-alerts e 2-critical e 3- errors © 4-warnings e 5-notifications © 6-informational 7-debugging You can filter log messages on the console by severity level by issuing the logging console severity- level command, or you can filter log messages to a Syslog server by issuing the logging trap severity- level command. When the logging console or logging trap command is issued with a severity-level parameter, messages with the specified severity level and all lower-numbered severity levels will be displayed or sent, respectively. Messages are formatted in the Berkeley Software Distribution (BSD) Syslog format, which is a percent sign (%) followed by a facility code, a severity code, and a mnemonic code. The three codes are separated by dashes. In the output displayed in this scenario, the facility code is DUAL, the severity code is 5, which is equivalent to notifications, and the mnemonic code is NBRCHANGE. The dash- separated code is followed by a colon and the human-readable text of the log message. The service timestamps debug datetime msec command, not the service timestamps debug datetime command, has been issued on RouterA. The service timestamps debug command can be used to configure the format in which the debug output will display the time. For example, issuing the service timestamps debug datetime command in global configuration mode will display a debug timestamp that consists of the date and time relative to Coordinated Universal Time (UTC) but will not include milliseconds. Issuing the service timestamps debug datetime msec command performs the same function as the service timestamps debug datetime command but adds the milliseconds value, which creates more granular timestamps and can thus increase the value of the logs following an attack. The output in this scenario includes a milliseconds value of .059. In contrast to the service timestamps debug command, you can issue the service timestamps log datetime msec command to display the date and time with milliseconds relative to UTC in logs that are generated by the Cisco device. Although debug messages on Cisco devices are displayed in the standard Syslog format, debug messages are not exactly the same as the log messages that are sent file send from Cisco device by Syslog to a or a server. You can Syslog messages a to a Syslog server by issuing the logging host ip-address command, where ip-address is the Internet Protocol (IP) address of the Syslog server. Syslog messages are typically sent over Transmission Control Protocol (TCP) port 514 or User Datagram Protocol (UDP) port 514. The default logging facility code is not being displayed in the output in this scenario. A logging facility, or facility code, is typically a free-form keyword value on Cisco devices that is used to identify the service or software that generated the message. For example, the output in this scenario has a facility code of DUAL. However, when a Cisco device sends Syslog messages to a Syslog server, a configured facility code is used to identify the device that sent the message. By default, Cisco devices are typically configured to use a value of local7. You can configure the logging facility that the device sends to the Syslog server by issuing the logging facility facility-type command, where facility-type is the facility code you want to send. There is not enough information in this scenario to determine whether the logging history size 1 command has been issued on RouterA. The logging history size value command, where value is a number from 0 through 500, can be issued to increase the size of a router's logging history table. The default history table size is 1, which means that only one log message is stored in the table at a time.

## Day 42 - SSH & Device Management

Included Exam C items: Q58, Q74

### Original Exam C Question 58

Which of the best describes should following statements why you use SSH to manage a Cisco device?  
(Select the best answer.)

- A. SSH uses the console port.

- B. SSH disables access to unused switch ports.
- C. SSH requires user names, whereas Telnet does not.
- D. \_ SSH is the only supported connection protocol.
- E. SSH provides encryption, whereas Telnet does not.

**Correct Answer: E**

#### Explanation

You should use Secure Shell (SSH) to manage a Cisco device because it provides encryption, whereas Telnet does not. SSH and Telnet are network protocols that are supported on Cisco devices and that are used to provide terminal emulation over virtual terminal (vty) lines. Cisco recommends using SSH rather than Telnet because SSH uses encryption. By contrast, Telnet sends traffic in clear text. Thus, when Telnet is used, an attacker could use a packet sniffer to view the contents of the Telnet traffic. When SSH is used, an attacker would not be able to view the contents of the management traffic. After implementing SSH, you can increase security by disabling Telnet access to the router. Both SSH and Telnet can be configured to use a user name/password combination. However, when SSH is used, user names and passwords are encrypted, whereas Telnet transmits them in clear text. Network security can be compromised if user names and passwords are not encrypted, because they can be intercepted and read by malicious users. Another security best practice is to disable access to unused ports. However, SSH cannot be used to accomplish this task. Instead, to disable access to unused ports, you can disable the unused ports with the shutdown command or you can create an isolated virtual local area network (VLAN) so that if malicious users gained access to the port, they would not be able access other portions of the network. SSH does not use the console port. Instead, SSH is typically used to connect over vty lines. The console port is a physical port that allows access to a router. Older Cisco devices have console ports that require a serial rollover cable; newer Cisco devices have console ports that support Universal Serial Bus (USB) connectivity.

### Original Exam C Question 74

You want to configure SSH for incoming vty connections on a new router. The router is running a K9 IOS image but has not yet been configured with a host name, a domain name, or an RSA key pair. In addition, the vty lines are not yet configured to accept incoming SSH connections.

You issue the `crypto key generate rsa` command from global configuration mode.

Which of the following messages will you most likely receive? (Select the best answer.)

- A. Please enable SSH as a transport mode. oO
- B. Please define a domain-name first.
- C. The name for the keys will be:
- D. Please define a hostname other than Router. oO
- E. Please create RSA keys to enable SSH.

**Correct Answer: D**

#### Explanation

You will most likely receive the Please define a hostname other than Router. message when you issue the `crypto key generate rsa` command, because you have not configured the router with a host name other than the default name of Router. To configure a router with a host name other than the default, you should issue the `hostname host-name` command from global configuration mode. To enable Secure Shell (SSH) for virtual terminal (vty) lines on a Cisco router, you should complete the following steps: 1. Configure the router with a host name other than Router by issuing the `hostname` command. 2. Configure the router with a domain name by issuing the `ip domain-name` command. 3. Generate an RSA key pair for the router by issuing the `crypto key generate rsa` command. 4. Configure the vty lines to use SSH by issuing the `transport input ssh` command from line configuration mode. SSH is often used as a secure replacement for Telnet to manage network devices. In order for SSH to be enabled on a Cisco device, the device must be running a K9 IOS image, which provides cryptographic functionality. You will not receive the The name for the keys will be: message when you issue the `crypto key generate rsa` command in this scenario. However, if you had already configured the router with a valid host name and a domain name, you would have received the The name for the keys will be: message after issuing the `crypto key generate rsa` command. After you specify the name for the keys, you will be prompted for the modulus length. You will not receive the Please define a domain-name first. message when you issue the `crypto key generate rsa` command in this scenario. However, if you had configured the router with a valid host name but had not configured the router with a domain name, you would have received the Please define a domain-name first. message after issuing the `crypto key generate rsa` command. In this scenario, you have configured neither the domain name nor the host name. To configure a router with a domain name, you should issue the `ip domain-name domain-name` command from global configuration mode. You will not receive the Please create RSA keys to enable SSH. message when you issue the `crypto key generate rsa` command in this scenario. However, if you had issued another command related to SSH, such as the `ip ssh time-out 60` command, but had not yet enabled SSH on the router, you would have received the Please create RSA keys to

enable SSH. message. You will not receive the Please enable SSH as a transport mode. message when you issue the crypto key generate rsa command in this scenario. The Please enable SSH as a transport mode. message is not a warning message that is displayed on Cisco routers. You can issue the transport input ssh command to configure SSH as the transport mode for vty lines.

## Day 43 - FTP & TFTP

Included Exam C items: Q55

### Original Exam C Question 55

Which of the following ports is used by TFTP? (Select the best answer.)

- A. TCP port 69
- B. TCP port 21
- C. UDP port 21
- D. UDP port 20
- E. TCP port 20

**Correct Answer: F**

Explanation

User Datagram Protocol (UDP) port 69 is used by Trivial File Transfer Protocol (TFTP). UDP is a Transport layer protocol that does not use sequence numbers or establish synchronized connections. Because of UDP's connectionless nature, transmitted datagrams can appear out of sequence or can be dropped without notice; thus it is the responsibility of the Application layer protocol to reorder packets or request the transmission of lost datagrams. TFTP is used to transfer files unreliably and without authentication over a network. Cisco devices can unreliably transfer IOS images by using TFTP. For example, administrators can use TFTP to transfer Cisco IOS images from a server to a device to perform firmware upgrades. TFTP does not use Transmission Control Protocol (TCP). TCP is a Transport layer protocol that uses sequencing and error-checking to ensure that transmitted data can be easily reordered if packets arrive out of sequence and can be retransmitted if any packets are lost. Because TCP handles data sequencing and the retransmission of lost data, the Application layer protocols that rely on TCP do not need to handle those tasks and can rely on receiving reliable, ordered data. File Transfer Protocol (FTP) uses TCP ports 20 and 21 to transfer files over a network. FTP uses TCP port 20 for data traffic and TCP port 21 for control traffic. Cisco devices can reliably transfer IOS images by using FTP. Unlike TFTP, FTP requires the transmission of authentication credentials, even if anonymous FTP is in use; however, those credentials are transmitted in plain text.

## Days 44-45 - NAT

Included Exam C items: Q69

### Original Exam C Question 69

{ Internet }

RouterA \ }

HostB HostA · · \_ ource Destination > ource Destination 192.168.1.22 | 192.168.1.30 15.16.17.18 Source Destination Source Destination 192.168.1.30 10 160122 | | 15.16.17.18 | dostA is a computer on your company's network. RouterA is a NAT router. HostA sends a packet to HostB, and HostB sends a packet back to HostA.

Which of the following addresses is an outside local address? (Select the best answer.)

- A. 22.23.24.25
- B. 15.16.17.18
- C. 192.168.1.30
- D. 192.168.1.22

**Correct Answer: C**

## Explanation

The 192.168.1.30 address is an outside local address. An outside local address is an Internet Protocol (IP) address that represents an outside host to the local network. Network Address Translation (NAT) translates between and IP addresses in order enable hosts addressed public private to on a privately network to access the Internet. Public addresses are routable on the Internet, and private addresses are routable only on internal networks. Several IP address ranges are reserved for private, internal use; these addresses, shown below, are defined in Request for Comments (RFC) 1918. © 10.0.0.0 through 10.255.255.255 © 172.16.0.0 through 172.31.255.255 © 192.168.0.0 through 192.168.255.255 The outside local address is often the same as the outside global address, particularly when inside hosts attempt to access resources on the Internet. However, in some configurations, it is necessary to configure a NAT translation that allows a local address on the internal network to identify an outside host. When RouterA receives a packet destined for 192.168.1.30, RouterA translates the 192.168.1.30 outside local address to the 15.16.17.18 outside global address and forwards the packet to its destination. To configure a static outside local-to-outside global IP address translation, you should issue the `ip nat outside source static outside-global outside-local` command. In this scenario, 15.16.17.18 is an outside global address. An outside global address is an IP address that represents an outside host to the global network. Outside global addresses are public IP addresses assigned to an Internet host by the host's operator. The outside global address is usually the address with the Domain host's IP address registered Name System (DNS) server to map a public to a friendly name such as `www.mycompany.com`. In this scenario, 192.168.1.22 is an inside local address. An inside local address is an IP address that represents an inside host to the local network. Inside local addresses are typically private IP addresses defined by RFC 1918. In this scenario, 22.23.24.25 is an inside global address. An inside global address is a publicly routable IP address that is used to represent an inside host to the global network. Inside global IP addresses are typically assigned from a NAT pool on the router. You can issue the `ip nat pool` command to define a NAT pool. For example, the `ip nat pool natpool 22.23.24.11 22.23.24.30 netmask 255.255.255.224` command allocates the IP addresses 22.23.24.11 through 22.23.24.30 to be used as inside global IP addresses. When a NAT router receives a packet destined for the Internet from a local host, it changes the inside local address to an inside global address and forwards the packet to its destination. In addition to configuring a NAT pool to dynamically translate addresses, you can configure static inside local-to-inside global IP address translations by issuing the `ip nat inside source static inside- local` command. This command `inside-global` maps a single inside local address on the local network to a single inside global address on the outside network. It is important to specify the inside and outside interfaces when you configure a NAT router. To specify an inside interface, you should issue the `ip nat inside` command from interface configuration mode. To specify an outside interface, you should issue the `ip nat outside` command from interface configuration mode. The following graphic depicts the relationship between inside local, inside global, outside local, and outside global addresses: ee { Internet j ) RouterA \ HostB HostA · Inside Local\_ | Outside Local Inside Global\_ | Outside Global Outside Local Inside Local Outside Global Inside Global | |

## Days 46-47 - QoS

Included Exam C items: Q9

### Original Exam C Question 9

Which of the following refers to the process of buffering packets in a software queue on a QoS-enabled interface? (Select the best answer.)

- A. scheduling
- C. traffic policing
- D. classification

**Correct Answer: B**

## Explanation

Queuing is the process of buffering packets in a software queue on a Quality of Service (QoS)- enabled interface. There are two types of queues on a QoS-enabled interface: a software queue and a hardware queue. The software queue can be configured to use various QoS mechanisms to control the flow of the the hardware transmit is packets through queue. By contrast, queue, or queue, a first- in-first-out (FIFO) queue that processes each packet without regard to its priority. All packets flow through the hardware queue before they are transmitted onto the physical medium. Typically, a packet is assigned to a particular traffic class and then placed into the corresponding queue until it can be transmitted. Classification is the process of dividing packets into categories based on predefined attributes, not the of in software When is received process buffering

packets a queue. a packet on a QoS-enabled interface, the QoS-enabled device inspects the packet's Data Link layer and Network layer traffic and then the into traffic class. Packets classified based descriptors places packet a are commonly on traffic descriptors such as application protocol or destination Internet Protocol (IP) address. Because each traffic class can be assigned its own queue, packets with different classifications are typically placed into separate queues. Scheduling is the process of handling packets in accordance with a particular QoS algorithm, not the process of buffering packets in a software queue. Packets that have been queued are transmitted in accordance with the scheduling method configured on the interface. For example, if an interface has been configured to use custom queuing (CQ), each queue can send a fixed number of bytes before the next queue can send traffic. Traffic policing is the process of regulating the rate of a particular traffic flow, not the process of buffering packets in a software queue. Traffic regulation techniques such as traffic policing and traffic shaping rely on the classification, queuing, and scheduling of packets in order to guarantee a level of service to certain classes of traffic. For example, traffic policing discards traffic flows that exceed the configured thresholds.

## Day 48 - Security Fundamentals

Included Exam C items: Q16, Q22, Q34, Q35, Q41, Q56, Q57, Q59, Q66

### Original Exam C Question 16

Which of the following types of malware can be used to modify the kernel of an OS and can be used by an attacker to access the computer in the future? (Select the best answer.)

- A. abackdoor
- B. aworm O spyware

**Correct Answer: D**

#### Explanation

A rootkit is a type of malicious software, or malware, that can be used to modify the kernel of an OS and can be used by an attacker to access the computer in the future. There are several types of rootkits, including kernel-level rootkits, hypervisor-level rootkits, library-level rootkits, and application-level rootkits. Modifying the kernel requires kernel-level privileges and can completely change the functionality of the infected system, potentially making the system vulnerable to other attacks. A rootkit can bypass the kernel by moving or attaching itself to the master boot record (MBR) so that it loads before the OS does. A rootkit can conceal malicious programs, such as keyloggers, sniffers, and other utilities, from antivirus and antimalware applications. A worm is a type of hidden software that can replicate itself over a network. Worms attempt to infect computers, replicate themselves, and write malicious code to applications on the infected computer. A worm can use network connections to automatically propagate without human interaction and infect other computers or use addresses found in the address book of an email client to send itself as an email attachment to other email addresses. A worm is not a type of malware that can be used to modify the kernel of an OS. Spyware is a type of malware that collects information and reports back to a third party without your permission. Spyware can record keystrokes, track your browsing habits, and steal information from your computer. Typically, you are unaware that spyware is being installed on your computer, because it can be installed along with other software you install. Spyware is not a type of malware that can be used to modify the kernel of an OS. A backdoor is a means by which an unauthorized user can obtain access to a network without being noticed by the network administrators. This is often accomplished by installing a program on a target computer. The backdoor typically circumvents the network authentication system, thus enabling the attacker to gain access to network resources. Although a backdoor can be used by an attacker to access a computer in the future, a backdoor is not used to modify the kernel of an OS.

### Original Exam C Question 22

Which of the following most accurately describes the difference between an NGFW and an NGIPS? (Select the best answer.)

- A. NGFW and NGIPS are different names for the same security appliance. An NGFW can target traffic based on source and destination addresses, whereas an
- B. can target traffic based on application types. An NGFW can filter traffic based on application types, whereas an NGIPS can block traffic based on application and user behavior. An NGFW can detect cyberattacks and block their traffic, whereas an NGIPS can target traffic based on source and destination addresses.

**Correct Answer: C**

## Explanation

Among the available choices, the most accurate statement is that a Next-Generation Firewall (NGFW) can filter traffic based on types of applications, whereas a Next-Generation Intrusion Prevention System (NGIPS) can block traffic based on application and user behavior. Traditional firewalls usually perform only stateless or stateful packet filtering. They target network traffic based on its direction, protocol, and source and destination IP addresses and ports. NGFWs can also attribute traffic to individual applications and can either allow or block traffic associated with specific applications or application categories. For example, an NGFW can be configured to allow traffic for collaboration applications and to block traffic for social networks. By contrast, an NGIPS can block traffic only if it presents a threat. An NGIPS can detect unusual threats by using signatures or by analyzing and identifying traffic anomalies. Although an NGFW is often integrated with an NGIPS, they are considered separate tools. The NGIPS capabilities of a Cisco NGFW are not enabled by default and require a separate subscription and license in order to be activated. An NGFW can target traffic based on source and destination addresses as well as types of applications. An NGIPS can detect cyberattacks and block their traffic but cannot target traffic based on source or destination addresses, unless those addresses are listed in its threat database as malicious. An NGFW cannot detect or block attacks on its own without using an NGIPS.

## Original Exam C Question 34

administrator issues the service command An password-encryption on a Cisco router.

Which of the following passwords will be encrypted on the router? (Select the best answer.)

- A. passwords configured by using the enable secret command only
- B. B. passwords configured by using the password command only passwords configured by any means along with any passwords configured in the future
- D. passwords configured by using the enable password command only

## Correct Answer: C

## Explanation

Passwords configured by any means along with any passwords configured in the future will be encrypted in this scenario because the administrator has issued the service password-encryption command. The service password-encryption command configures global password encryption in a running configuration. Any passwords that were not previously encrypted will be encrypted. In addition, any passwords that are created in the future, regardless of the commands used to create them, will be encrypted in the running configuration. Passwords configured by using the enable secret command are already encrypted and will be encrypted in the running configuration when the service password-encryption command is issued the Cisco versions the enable command by default on router. In some IOS prior to 15.3(3), secret stores an encrypted password in the device's configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. The syntax for the enable secret command is enable secret [level level] {password | [encryption-type] encrypted-password}, where password is a string of characters that represents the clear-text password. Instead of supplying a clear-text password, you can specify an encryption-type value of 0, 4, or 5 and an encrypted-password value of either a clear-text password, a SHA-256 hash, or a Message Digest 5 (MD5) hash, respectively. Supplying a hash value requires that you have previously encrypted the value by using a hashing algorithm in the same fashion that IOS uses the algorithm. This command configures a password that is required in order to place the device into enable mode, which is also known as privileged EXEC mode. The device must, at a minimum, be placed into enable mode for the user to be able to display the running configuration. Passwords configured by using the enable password command are not encrypted by default but will be encrypted in the running configuration when the service password-encryption command is issued on the router. You can issue the enable password command to create a password that must be used to gain access to enable mode. The syntax of the enable password command is enable password [level level] {password | [encryption-type] encrypted-password}. Thus issuing the enable password level 7 cisco command would configure cisco as the password required to access privilege level 7. However, unlike the enable secret command, the enable password command would configure the password to be stored as clear text. If both the enable password command and the enable secret command are in the running configuration of a Cisco device, the device will ignore the password associated with the enable password command. If all passwords on the device have been encrypted by issuing the service password-encryption command, the router will still prefer the enable secret password over the enable password password. Passwords configured by using the password command are not encrypted by default but will be encrypted in the running configuration when the service password-encryption command is issued on the router. The line console 0 command followed by the password command configures a password for accessing the router by using the console. Typically, the console is accessed by physically connecting a console cable between the router and a device that is running terminal software. Issuing the password for the console places the device into user mode.

## Original Exam C Question 35

Which of the following is an example of authentication by something you have? (Select the best answer.)

- A. a password
- B. a PIN or your fingerprints

**Correct Answer: C**

### Explanation

Your driver's license is an example of authentication by something you have. There are three typical methods of authentication for gaining access to a secure environment: something you know, something you have, and something you are. A fourth possible method is authentication by someplace you are, which means that you can be granted access to a secure system by virtue of your workstation's location on a network, such as an Internet Protocol (IP) address or your physical location in the world. Authentication by one type of factor, such as something you know, is known as single-factor authentication. Authentication by more than one type of factor, such as something you know and something you are, is known as multifactor authentication. However, requiring more than one of any single factor, such as two knowledge factors, is not considered multifactor authentication. Authentication by something you have is the process of verifying your identity by using a device or document that with such driver's mobile you carry you, as a fob, a license, a smart card, or a phone with an authenticator application. For example, a police officer who stops you can verify your identity by comparing an image on your driver's license to your physical appearance. In addition, your company might require that you carry a fob in order to gain access to the office by using an exterior door. You would typically hold the fob in front of a sensor and allow the sensor to read the information that is stored on the fob; if the information matches that of an authorized user, you would be allowed entry. Authentication by something you have is also known as Type 2 authentication. Authentication by something you have is considered a stronger form of authentication than authentication by something you know because it requires the user to carry some sort of authenticating electronic access control (EAC) token. A password or a personal identification number (PIN) is an example of authentication by something you know. For example, a bank's website might choose to ask you to provide both a password and the answer to a security question such as your mother's maiden name. Although the bank's website prompts you for two forms of verification, both of those prompts are for information that you store in your memory. Authentication by something you know is also known as Type 1 authentication. Authentication by something you know is considered the weakest form of authentication because such authentication can often be easily guessed or broken by brute force. Fingerprints are an example of authentication by something you are. Authentication by something you are is the process of verifying your identity by using something that is unique about you and that cannot be easily changed, such as your iris, your retina, or your fingerprints. For example, your company could connect your workstation to a fingerprint scanner instead of requiring you to unlock your workstation with keystrokes and a password. Authentication by something you are is also known as Type 3 authentication. Authentication by something you are is considered the strongest form of authentication because of the unique biometrics of individuals.

## Original Exam C Question 41

Which of the following are most likely to be considered forms of authorization? (Select 2 choices.)

- D. A. assigning a role to a verified user
- B. verifying a user's fingerprint pattern
- C. verifying a user's password
- D. logging a verified user's file access

**Correct Answer: A, E**

### Explanation

Assigning a role to a verified user and allowing a user to access a specific file are forms of authorization. Authentication, Authorization, and Accounting (AAA) systems manage user activity. AAA systems are typically more sophisticated than simple password authentication systems, such as a local password database. Allowing access to a specific file can involve assigning specific user or group permissions directly to the file, matching a rule, such as an access control list (ACL) that associates a specific user with a file, or assigning a user to a specific role that has permission to access the file. Two common AAA systems are Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access-Control System Plus (TACACS+). RADIUS is a standard AAA protocol created by the Internet Engineering Task Force (IETF). Compared to TACACS+, RADIUS has several limitations. For example, RADIUS encrypts only the password in Access-Request packets; it does not encrypt the entire contents of the packet like TACACS+ does. TACACS+ is a Cisco-proprietary protocol used during AAA operations. TACACS+ provides more granular and flexible control over user access privileges. For example, the AAA

operations are separated by TACACS+, whereas RADIUS combines the authentication and authorization services into a single function. Because TACACS+ separates these functions, administrators have more control over access to configuration commands. In addition, TACACS+ encrypts the entire contents of packets, thus providing additional security. The following list defines the three phases of the AAA process:

- Authentication · the process of verifying a user's identity
- Authorization · the process of verifying the level of access configured for a user
- Accounting · the process of recording the use of resources

Verifying a user's fingerprint pattern and verifying a user's password are both likely to be considered authentication, not authorization. Authentication is the process of verifying a user's identity. Authentication by itself does not grant access to a given resource. Logging a verified user's file access is a form of accounting, not authorization. Accounting is a feature of AAA systems that enables administrators to track resource usage across a network. If a security incident occurs, accounting can aid the effort to track the incident back to its source.

## Original Exam C Question 56

Drag each IOS command to its description. Commands can be used only once. Use all commands.

Descriptions include: configure and encrypt a clear-text enable password; configure an encrypted vty login password; enable global password encryption; configure a clear-text enable password; configure a previously encrypted enable password.

**Correct Answer: enable secret password -> configures and encrypts a clear-text password for enable mode**  
**password 7 hash -> configures an encrypted vty login password**  
**service password-encryption -> enables global password encryption**  
**enable password password -> configures a clear-text password for enable mode**  
**enable secret 5 hash -> configures a previously encrypted password for enable mode**

### Explanation

The enable secret password command configures and encrypts a clear-text password for gaining access to enable mode. The enable secret 5 hash command, on the other hand, configures a previously Message Digest 5 (MD5)-encrypted password for gaining access to enable mode. In some Cisco IOS versions prior to 15.3(3), the enable secret command by default stores an encrypted password in the device's configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. The syntax for the enable secret command is enable secret [level level] {password | [encryption-type] encrypted-password}, where password is a string of characters that represents the clear-text password. Instead of supplying a clear-text password, you can specify an encryption-type value of 0, 4, or 5 and an encrypted-password value of either a clear-text password, a SHA-256 hash, or an MD5 hash, respectively. Supplying a hash value requires that you have previously encrypted the value by using a hashing algorithm in the same fashion that IOS uses the algorithm. This command configures a password that is required in order to place the device into enable mode, which is also known as privileged EXEC mode. The device must, at a minimum, be placed into enable mode for the user to be able to display the running configuration. The enable command clear-text enable password password configures a password on a Cisco device. If both the enable password command and the enable secret command are in the running configuration of a Cisco device, the device will ignore the password associated with the enable password command. The service password-encryption command enables global password encryption on a Cisco device. When service password-encryption is issued, any passwords that are currently configured as clear-text in the running configuration will be encrypted. In addition, any future passwords created in the configuration will be encrypted. However, even if all passwords on a device have been encrypted because the service password-encryption command has been configured, as in this scenario, the router will still prefer the enable secret password over the enable password password. The password 7 hash command configures an encrypted virtual terminal (vty) login password when the command is issued in vty line configuration mode. For example, you can configure Telnet login information on a Cisco device by issuing the line vty first last command to place the device in vty line configuration mode. Next, you can issue the password password command to configure a Telnet password and the login command to enable password checks if the command has been disabled on the router. The login command is typically configured by default. Issuing the Telnet password when you are connecting to a device places the device into user EXEC mode, where it is not typically possible to display the running configuration.

## Original Exam C Question 57

named boson with the The You want to create a user account password eX\$1mM0x on a router. password should be converted to an MD5 hash and stored on the router.

Which of the following commands should you issue on the router? (Select the best answer.)

- A. username boson password eX\$1mM0x
- B. username boson secret 5 eX\$1mM0x
- D. username boson eX\$imM0x

## Correct Answer: C

### Explanation

To create a user account named boson with a Message Digest 5 (MD5)-hashed password of eX\$1mM0x, you should issue the `username boson secret eX$1mM0x` command on the router. The `username` command creates a new user and adds the user to the local user database on a router. The local database contains user on a router a list of users that have been added to the router; these users can access the router. When using the `username` command to create a new user on a router, you can configure the user's password to be stored as plain text or as an MD5 hash. To configure a user name with a plain text password, you should use the `username user-name password password` command. Using the `secret` keyword instead of the `password` keyword ensures that the password is stored as an MD5 hash. Thus the command `username boson secret eX$1mM0x` creates a user named boson and stores the password as an MD5 hash value. In the output of the `show running-config` command, the hash value of the password rather than the actual password would be displayed, similar to the following: `username boson secret 5 $%A*mNXyz001976gtr` The 5 indicates that the password was encrypted with MD5. Issuing the `username boson password eX$1mM0x` command creates the boson user account and adds the user account to the local user database. However, the password is stored as plain text instead of an MD5 hash because the `password` keyword is used instead of the `secret` keyword. If you know the hash value of the password, you can use the MD5 hash value of a password manually instead of assigning a plain text password to be converted into a hash by the IOS. This is accomplished by issuing the `username user-name secret 5 hash-value` command. The 5 parameter indicates that the assigned value is already in MD5 hash form. The scenario indicates that the password should be converted to an MD5 hash, so you should not issue the `username boson secret 5 eX$1mM0x` command. The `username boson eX$1mM0x` command is an invalid Cisco command because it does not contain the `password` keyword. Either `password` or `secret` is required when issuing the `username` command.

## Original Exam C Question 59

You issue the following command set on a Cisco router:

```
enable secret BO$On!
```

```
line vty 04
```

```
password C1$c0!
```

```
login
```

```
service password-encryption
```

Which of the following password types is the BO\$On! password stored as in the running configuration?

(Select the best answer.)

A. Type0

B. Type7

C. Types

E. Typed

## Correct Answer: D

### Explanation

The BO\$On! password will be stored as a Type 5 password because the `enable secret BO$On!` command has been issued in this scenario. The `enable secret` command by default stores a password in Message Digest 5 (MD5)-encrypted format in the device's running configuration. In some Cisco IOS versions prior to 15.3(3), the `enable secret` command by default stores an encrypted password in the device's running configuration file by using a Secure Hash Algorithm (SHA) 256-bit hash. As of Cisco IOS 15.3(3), Type 4 passwords have been deprecated because of a security flaw in their implementation. Similar to the `enable secret` command, the `username user-name secret password` command encrypts the password associated with the given user as a Type 5 password in the running configuration. It is important to note that if both the `enable password password` command and the `enable secret password` command are configured on the same device, the device will prefer the Type when is for the enable mode 5 password a user prompted password. The BO\$On! password will not be stored as a Type 0 password in this scenario. Type 0 passwords are passwords that are stored without encryption in the running configuration. Some examples of Cisco commands that, by default, store passwords as Type 0 passwords are the `enable password password` command, the `line configuration mode password password` command, and the `global configuration mode username user-name password password` command. The BO\$On! password will not be stored as a Type 7 password in this scenario. However, the C1\$c0! password will be stored as a Type 7 password in this scenario because the `service password- encryption` command has been issued. When you issue the `service password-encryption` command on a Cisco device, all current and future Type 0, or clear-text, passwords are encrypted as Type 7 passwords in the device's running configuration. Because a password that is configured by using the `enable secret` command is already encrypted as a Type 5 password in the running configuration, that password will not be re-encrypted as a Type 7 password by the `service password-encryption` command. Type 7 passwords are

encrypted by using the Cisco-proprietary Vigenère encryption algorithm. This type of encryption prevents anyone who views the device's running configuration from easily reading the passwords that are configured there. However, Type 7 encryption is weak. Type 7 passwords can be easily decrypted by using tools that are freely available online. Where possible, you should use a stronger encryption method to encrypt passwords in the running configuration. There is nothing in this scenario to indicate that the BO\$On! password will be stored as a Type 8 password. Type 8 passwords are encrypted by using a different implementation of SHA-256 than the Type 4 implementation of SHA-256. Type 8 passwords are encrypted by using a Password-Based Key Derivation Function 2 (PBKDF2) with a SHA-256 secret. Type 8 passwords are not reversible and are considered to be stronger than Type 4, Type 5, and Type 7 passwords. You can configure a Cisco device to use Type 8 passwords by default when you issue the enable secret password command. To configure Type 8 passwords as the default secret type, issue the enable algorithm-type sha256 command in Cisco IOS versions later than 15.3(3)M3. There is nothing in this scenario to indicate that the BO\$On! password will be stored as a Type 9 password. Type 9 passwords are encrypted in the running configuration by using the scrypt encryption algorithm. Like Type 8 passwords, Type 9 passwords are not reversible. You can configure a Cisco device to use Type 9 passwords by default when you issue the enable secret password command. To configure Type 9 passwords as the default secret type, issue the enable algorithm-type scrypt command in Cisco IOS versions later than 15.3(3)M3. It is important to note that if you plan to downgrade a device to a Cisco IOS version earlier than 15.3(3)M3 and you have configured either Type 8 or Type 9 encryption, you should first issue the enable algorithm-type md5 command in 1OS 15.3(3)M3 and reconfigure the enable secret password command. Otherwise, you will be locked out of the Cisco device after downgrading.

### Original Exam C Question 66

Which of the following IDS devices are the best at detecting new security threats? (Select the best answer.)

- A. signature-matching IDSs
- B. pattern-matching IDSs
- C. anomaly-detection IDSs
- D. protocol-behavior IDSs

**Correct Answer: C**

#### Explanation

Anomaly-detection intrusion detection systems (IDSs) are the best at detecting new security threats. Anomaly-detection devices use machine learning and artificial intelligence (AI) to detect abnormal patterns in network traffic behavior. To enable anomaly-detection devices to detect abnormalities in traffic, the devices must first take a baseline reading of what normal network traffic patterns are like. Once the baseline is taken, an anomaly-detection device will compare future traffic against the baseline to detect abnormal traffic flows. Although anomaly-detection devices are good at detecting new attacks, they have a higher false positive rate. Pattern-matching IDSs, which are also called signature-matching IDSs, use specific strings of text called signatures to detect malicious traffic. The primary benefit of signature-matching detection methods is that the number of false positives generated is typically low. However, signature-matching detection methods do not provide adequate protection against new attacks. Although signatures can be added threats there is between the time as new are found, always a delay a threat is released and the time a signature is developed to detect the threat. Protocol-behavior IDSs use rules to detect protocol traffic that does not follow standard methods of operation. The rules used by protocol-behavior devices are usually based on the Request for Comment (RFC) documents that define each protocol. Although protocol-behavior devices can detect nonstandard traffic, there is no way to know for sure whether the traffic is caused by a malicious user or by a poorly coded application. Therefore, protocol-behavior devices have a higher rate of false positives and they are not as good as anomaly-detection devices at detecting new threats.

## Day 49 - Port Security

Included Exam C items: Q12, Q62

### Original Exam C Question 12

Which of the following commands should you issue on a switch port so that no more than two devices can send traffic into the port? (Select the best answer.)

- O switchport port-security mac-address sticky
- C. switchport port-security mac-address 2

- D. switchport port-security 2
- E. switchport port-security

**Correct Answer: A**

#### Explanation

You should issue the switchport port-security maximum 2 command from interface configuration mode so that only two devices can send traffic into the port. The switchport port-security maximum 2 command configures the switch port to allow no more than two devices, each with a unique Media Access Control (MAC) address, to send traffic into the port. A switch builds its MAC address table by analyzing the MAC addresses of ingress, or inbound, traffic on each port. Port security can be used to limit the number of MAC addresses that can be learned on each port. Port security allows traffic into a switch port from authorized MAC addresses. If traffic arrives from a MAC address that is authorized, the traffic will be forwarded to its destination. If traffic arrives from a MAC address that is not authorized, the traffic will be discarded, and on some switch configurations, the switch port will also be disabled. By itself, the switchport port-security command enables port security and authorizes a maximum of one MAC address to send traffic into the port. The 2 parameter in the switchport port-security 2 command is an invalid parameter; thus this command will not allow two devices to communicate on a switch port. Authorized MAC addresses can be statically configured or dynamically learned. To statically configure a switch port to allow traffic from a MAC address, you should issue the switchport port-security mac-address mac-address command in interface configuration mode. The 2 parameter in the switchport port-security mac-address 2 command is an invalid parameter; thus this command will allow devices communicate not two to on a switch port. Any MAC addresses that are not configured statically will be learned dynamically from incoming traffic, up to the maximum number of MAC addresses configured in the switchport port-security maximum number command. The switchport port-security mac-address sticky command converts dynamically learned MAC addresses to sticky MAC addresses. Sticky MAC addresses are stored in the running configuration. To ensure that the sticky MAC addresses are not lost during a reboot, you should issue the write memory or copy running-config startup-config commands. You cannot use the switchport port-security mac-address sticky command by itself to authorize a maximum of two devices to send traffic into a switch port; you must also issue the switchport port-security maximum 2 command.

#### Original Exam C Question 62

You issue the following commands on

```
SwitchA#configure terminal
```

```
SwitchA(config)#interface fastethernet 0/1
```

```
SwitchA(config-if)#switchport port-security
```

```
SwitchA(config-if)#switchport port-security maximum 12
```

You want to configure SwitchA to discard traffic and increment the SecurityViolation counter when it receives traffic on FastEthernet 0/1 from a host with an unauthorized MAC address.

Which of the following commands should you issue? (Select the best answer.)

- B. switchport port-security violation protect O switchport port-security violation shutdown O switchport port-security violation discard

**Correct Answer: A**

#### Explanation

You should issue the switchport port-security violation restrict command to configure SwitchA to discard traffic and increment the SecurityViolation counter when SwitchA receives traffic on FastEthernet 0/1 from a host with an unauthorized Media Access Control (MAC) address. The syntax of the switchport port-security violation command is switchport port-security violation {protect | restrict | shutdown}. The switchport port-security violation protect command configures a switch port to discard traffic that it receives from unauthorized hosts. However, the SecurityViolation counter is not incremented when the protect keyword is used. The switchport port-security violation shutdown command configures a switch port to enter the err-disabled state when the port receives traffic from unauthorized hosts. You can remove the switch port from the err-disabled state by issuing the errdisable recovery cause shutdown command from global configuration mode or by issuing the shutdown and no shutdown commands from interface configuration mode. The switchport port-security violation discard command contains incorrect syntax, because discard is not a valid keyword of the switchport port-security violation command.

## Day 51 - Dynamic ARP Inspection

Included Exam C items: Q37

### Original Exam C Question 37

Router1 Switch1 DHCP Server Host1 Host2 Host3 Host?

Host4 HostS Host6 You are configuring security for the network shown in the topology diagram. You want to configure DAI on Switch1.

Which port or ports on the switch should you configure as trusted? (Select the best answer.)

- A. all of the ports
- C. ports Fa1/1, Fa1/2, Fa1/3, and Fa1/4
- D. none of the ports

**Correct Answer: B**

Explanation

You should configure port Fa2/1 on Switch1 as a trusted port when you are configuring Dynamic ARP Inspection (DAI) on the switch. DAI can be used to help mitigate Address Resolution Protocol (ARP) poisoning attacks, which are also known as ARP spoofing attacks. In an ARP poisoning attack, an attacker intercepts an ARP request packet and replies with the attacker's own Media Access Control (MAC) address, rather than the address of the intended recipient. Subsequently, the attacker can intercept any traffic intended for the original recipient. DAI can be configured on individual ports or on entire virtual local area networks (VLANs). DAI can be enabled on a single VLAN or on multiple VLANs. To enable DAI, you use the `ip arp inspection vlan` global configuration command. The syntax of the `ip arp inspection vlan` command is `ip arp inspection vlan {vlan-ID | vlan-range}`. A range of VLANs can be entered by using a comma-separated list and/or a dash-separated pair of VLAN numbers indicating the range of VLANs to include. For example, each of the following commands enables DAI on VLANs 2 through 4: `ip arp inspection vlan 2,3,4` and `ip arp inspection vlan 2-4`. Configuring DAI on each VLAN ensures that traffic sent from each host is inspected. In addition, each port is by default an untrusted port. Issuing the `ip arp inspection` command in interface configuration mode enables DAI on an interface and by default configures the interface as an untrusted port. When configured, the interface is labeled either trusted or untrusted. DAI configured on a switch, ports are as follows: An ARP packet that is received on a trusted port is not inspected, whereas a packet received on an untrusted port is inspected. If a packet received on an untrusted port matches a valid entry in the switch's ARP table, the packet is forwarded; otherwise, the packet is dropped. By default, all ports are untrusted when DAI is configured. However, interfaces that connect to other routers or switches, such as the Fa2/1 interface, should be configured as trusted by using the `ip arp inspection trust` command. When DAI is configured for an entire VLAN, you can override the default configuration for a given port by issuing the `ip arp inspection trust` command in interface configuration mode. Ports Fa1/1, Fa1/2, Fa1/3, and Fa1/4 should not be configured as trusted ports. Because they are connected to host computers, they should be untrusted. If a port is configured as an untrusted port, ARP replies that originate from hosts connected to the port will be inspected; only ARP replies matching valid entries in the switch will be permitted on untrusted ports.

## Day 52 - LAN Architectures

Included Exam C items: Q51

### Original Exam C Question 51

In a two-tier network design, which layers are combined together? (Select the best answer.)

- A. distribution and access
- B. aggregation and access
- C. core and distribution
- D. core, aggregation, and access
- E. core and access
- F. core, distribution, and access

**Correct Answer: C**

Explanation

In a two-tier network design, the core and distribution layers are combined together into a single layer; this two-tier network design is often called the collapsed core design. The functionality of the core layer can be collapsed into the distribution layer if the distribution layer infrastructure is sufficient to meet the design requirements. The collapsed core in a two-tier network design provides physical and logical paths as well as a Layer 2 aggregation and demarcation point. In addition, a collapsed core defines routing policies and network access policies and provides intelligent network services. The Cisco hierarchical network model divides the network into three distinct layers: © Core layer © Distribution layer, sometimes called the aggregation layer © Access layer The core layer of the hierarchical model is primarily associated with low latency and high reliability. As the network backbone, the core layer provides fast convergence and typically provides the fastest switching path in the network. The distribution layer provides route filtering and interVLAN routing. The distribution layer serves as an aggregation point for access layer network links. Because the distribution layer is the intermediary between the access layer and the core layer, the distribution layer is the ideal place to enforce security policies, to provide Quality of Service (QoS), and to perform tasks that involve packet manipulation, such as routing. Summarization and next-hop redundancy are also performed in the distribution layer. The access layer serves as a media termination point for endpoints such as servers and hosts. Because access layer devices provide access to the network, the access layer is the ideal place to perform user authentication.

## Day 53 - WAN Architectures

Included Exam C items: Q53

### Original Exam C Question 53

Which of the following is true about IPSec configured in transport mode? (Select the best answer.)

- A. It does not encrypt the IP header.
- B. It encrypts the entire packet.
- C. It requires additional headers.
- D. It is required for NAT traversal.

**Correct Answer: A**

#### Explanation

Internet Protocol Security (IPSec) configured in transport mode does not encrypt the Internet Protocol (IP) header. IPSec is a suite of protocols that can be used to encrypt Generic Routing Encapsulation (GRE) tunnel traffic, such as over a virtual private network (VPN). IPSec supports two modes: transport and tunnel. In transport mode, only the IP packet's payload is encrypted by IPSec, which means that the IP packet's header remains intact. IPSec transport mode is not required for Network Address Translation (NAT) traversal. In fact, IPSec in transport mode is less compatible with NAT than IPSec when configured in tunnel mode. IPSec in tunnel mode encrypts the entire packet, including both the header and payload. The encrypted packet is then encapsulated in a new IP packet that includes a new IP header, which can be used by NAT. Transport mode, on the other hand, creates complications for NAT if Authentication Header (AH) is used. It is possible to use IPSec in transport mode when NAT is deployed. However, it requires the use of a NAT Traversal (NAT-T) solution as described in the Internet Engineering Task Force (IETF) Request for Comments (RFC) 3947. IPSec in tunnel mode, not in transport mode, requires additional headers. In addition, IPSec in tunnel mode, not in transport mode, encrypts the entire packet. Tunnel mode requires additional headers because the entire packet is encrypted. Layer 3 forwarding devices are not capable of decrypting the encrypted packet. Therefore, a new unencrypted IP header encapsulates the encrypted packet for routing.

## Day 55 - Wireless Fundamentals

Included Exam C items: Q64

### Original Exam C Question 64

You administer the network of a manufacturing plant in New York City. You want to install three 802.11g wireless APs in the administrative offices.

Which channels should you configure on the three APs? (Select the best answer.)

- A. 11
- B. B. any channel, as long as each AP uses a different channel
- C. 1,2,and3
- D. any channel, as long as all three APs use the same channel O. 1,7,and 13

**Correct Answer: A**

#### Explanation

You should configure one access point (AP) to use channel 1, one AP to use channel 6, and one AP to use channel 11 within the 2.4-gigahertz (GHz) frequency band. Although 802.11g can be configured to use 11 different channels in the United States and Canada, only three nonoverlapping channels can be used: 1, 6, and 11. Neighboring APs should use nonoverlapping channels so that the APs do not interfere with one another. Therefore, you should not use the same channel for all of the APs, nor should you use channels that overlap, such as channels 1, 2, and 3. The following table displays the 802.11g channels that can be used in the United States and Canada, along with their respective frequency ranges: Range (GHz) 2.406 - 2.428 2.411 - 2.433 2.416 - 2.438 2.421 - 2.443 2.431 - 2.453 2.436 - 2.458 2.441 - 2.463 2.446 - 2.468 You cannot use channels 1, 7, and 13 for the wireless network, because channel 13 cannot be used in the United States. Channels 1 through 13 can be used on 802.11g wireless networks in Europe and Japan.

## Day 56 - Wireless Architectures

Included Exam C items: Q30, Q32

### Original Exam C Question 30

Which of the following WLC interfaces supports mobility management for wireless clients? (Select the best answer.)

- A. the service port interface
- C. the management interface
- D. adynamic interface
- E. the AP-manager interface

**Correct Answer: B**

#### Explanation

The virtual interface on a wireless LAN controller (WLC) supports mobility management for wireless clients. A WLC can contain up to four static interfaces: the management interface, the AP-manager interface, the virtual interface, and the service port interface. The virtual interface supports mobility management by providing a specific Internet Protocol (IP) address that is the same across multiple controllers when wireless clients roam among the controllers. This enables seamless roaming among the controllers. The virtual interface supports embedded Layer 3 security. For example, the virtual interface can also be used in situations where web authorization has been enabled for clients; the user is redirected to the IP address of the virtual interface when the user opens a web browser. In addition, if Dynamic Host Configuration Protocol (DHCP) relay has been enabled on the controller, the virtual interface can be used as the DHCP server address on wireless clients. The AP-manager interface contains the IP address that is used as the source IP address by which the lightweight access points (APs) communicate with the WLC. Because the AP-manager interface communicates with the lightweight APs on the wireless network, the IP address assigned to the AP- manager interface should be unique on the network. After the interface has been configured, the WLC uses the AP-manager interface to listen for Layer 3 Lightweight Access Point Protocol (LWAPP) communications. The management interface is used for in-band management information. This interface is used for all Layer 2 LWAPP communications between the controller and the lightweight APs. In addition, the management interface is used to communicate with other WLCs on the wireless network. The service port interface is used for out-of-band maintenance purposes on a WLC. This interface is a physical interface on the WLC that can be used to recover the WLC in the event that the WLC fails. The service port can be configured with an IPv4 or IPv6 address and accessed using Telnet or Secure Shell (SSH). The service port interface is the only interface that is available while the WLC is booting. In addition to the four static interfaces, a WLC can contain up to 512 dynamic interfaces. Dynamic interfaces are user-defined and are typically used for wireless client data. The dynamic interfaces function similarly to virtual local area networks (VLANs). For example, you can create a dynamic interface to segment traffic on the WLC.

## Original Exam C Question 32

Which of the following best describes a lightweight AP in bridge mode? (Select the best answer.)

- A. It captures wireless traffic for analysis.
- B. It is the default operating mode for a lightweight AP.
- C. It acts as a dedicated connection between two networks.
- D. It enables a failsafe if the CAPWAP connection goes down.

**Correct Answer: C**

### Explanation

A Cisco lightweight access point (AP) operating in bridge mode acts as a dedicated connection between two networks. A lightweight AP provides an interface for wireless clients to connect to the wireless local area network (WLAN). However, unlike autonomous APs, a lightweight AP relies on a Cisco wireless LAN controller (WLC) for management and configuration. Lightweight APs operating in bridge mode can connect to other networks in either a point-to-point or point-to-multipoint fashion. When multiple APs are configured in bridge mode, the collection of lightweight APs can be used to form a mesh network, which increases reliability. Local mode is the default operating mode for a lightweight AP. A Cisco lightweight AP operating in local mode is capable of providing multiple basic service sets (BSSs) on a single channel. In this mode, the AP can connect to a WLC and can provide client connectivity. In addition, an AP operating in local mode scans all wireless channels as a means of monitoring wireless quality and security. The connection between and is created tunnels established by the a lightweight AP a WLC by using two Control and Provisioning of Wireless Access Points (CAPWAP) tunneling protocol. Information sent between lightweight APs and the WLC is encapsulated in Internet Protocol (IP) packets. This process enables a lightweight AP and the WLC to manage connectivity to the same WLAN yet be separated by both physical and logical means. Because the round-trip latency between the AP and the WLC cannot exceed 20 milliseconds, Cisco recommends that an administrator configure the physical network interface on a local mode AP to operate in access mode and enable the PortFast feature on that interface in order to mitigate delays. A Cisco lightweight AP operating in sniffer mode, not bridge mode, captures wireless traffic for analysis. When traffic is captured, a lightweight AP that is operating in sniffer mode will send the traffic to an analyzer, which is typically software that is installed on a PC or other host. A Cisco lightweight AP operating in FlexConnect mode, not bridge mode, enables a failsafe if the CAPWAP connection goes down. FlexConnect mode does not provide BSSs. When configured, FlexConnect mode enables a lightweight AP to switch traffic between a given Service Set Identifier and (SSID) a given virtual local area network (VLAN).

## Day 57 - Wireless Security

Included Exam C items: Q25, Q40

## Original Exam C Question 25

Which of the following can be implemented to help protect against spoofed management frames in a wireless network? (Select the best answer.)

- A. 802.11w
- B. 802.11v
- C. 802.11r
- D. 802.11k

**Correct Answer: A**

### Explanation

The Institute of Electrical and Electronics Engineers (IEEE) 802.11w standard can be implemented to help protect against spoofed management frames in a wireless network. 802.11w provides management frame protection (MFP) in a wireless local area network (WLAN). MFP is a security mechanism that prevents management frames from being compromised by a malicious user. Management frames are used to manage the connection between an access point (AP) and a wireless client. Numerous types of management frames exist: beacons, probe requests, probe responses, association requests, association responses, authentication requests, authentication responses, deauthentications, reassociation requests, reassociation responses, and announcement traffic indication messages. Each of these frame types performs specific duties necessary to

initialize, terminate wireless transmissions between and maintain, or an AP a client. Because management frames are not encrypted, they are frequently targeted by malicious users who want to disrupt network access via a Denial of Service (DoS) attack. In a DoS attack, an attacker floods a host with packets in an attempt to overload the host so that it cannot respond to further requests. The IEEE 802.11r standard provides fast transition roaming, not protection against spoofed management frames, on a wireless network. With 802.11r, a roaming wireless client and an AP perform the Pairwise Master Key (PMK) calculation in advance. These precalculated PMK keys are then used after the client sends its reassociation request to the new AP and the AP responds with its reassociation response. This process enables the client to transition to a new AP without needing to reauthenticate, thereby speeding the handoff between APs and minimizing adverse Quality of Service (QoS) effects caused by roaming. The IEEE 802.11k standard provides assisted roaming, not protection against spoofed management frames, on a wireless network. With 802.11k, a roaming wireless client can request a list of known neighboring APs that might be potential candidates for transition. The list returned by an AP is optimized for the client and contains only APs in the same wireless band with which the client is associated. This enables the client to find a new AP without needing to probe unnecessary wireless channels, which speeds transition and increases battery life for portable devices. The IEEE 802.11v standard provides network-assisted power savings, not protection against spoofed management frames, on a wireless network. With 802.11v, a wireless client can remain in a standby or sleep state longer to improve battery life. 802.11v provides a Directed Multicast Service (DMS), which enables a client to request that certain multicast messages be delivered as unicast messages instead. This enables the client to ignore the multicast messages during sleep and to receive the unicast messages at a greater rate of speed once the client is active again. In addition, 802.11v provides a client with the ability to extend the amount of time it may remain idle before it is disassociated with an AP. This enables the client to wake less frequently from sleep and extends the life of the client's battery.

### Original Exam C Question 40

Which of the following technologies is implemented in WPA3-Personal to replace the use of PSKs in WPA2- Personal? (Select the best answer.)

- A. ECDSA
- B. AES-GCMP
- D. PMF

**Correct Answer: C**

Explanation

Of the available choices, Simultaneous Authentication of Equals (SAE) is the technology that is implemented in Wi-Fi Protected Access 3 (WPA3)-Personal to replace the use of pre-shared keys (PSKs) in WPA2-Personal. WPA3 was introduced by the Wi-Fi Alliance in 2018 to address weaknesses in WPA2. SAE enables clients and access points (APs) to mutually authenticate. WPA3-Personal introduces support for Protected Management Frames (PMF). However, PMF does not replace PSKs. PMF secures management frame communication between an access point (AP) and a client. WPA3-Enterprise, not WPA3-Personal, improves authentication encryption by introducing support for 256-bit Advanced Encryption Standard (AES)-Galois/Counter Mode Protocol (GCMP). Key establishment and authentication is accomplished by using Elliptic Curve Diffie-Hellman (ECDH) Exchange and Elliptic Curve Digital Signature Algorithm (ECDSA) with a 384-bit curve.

## Day 61 - REST APIs

Included Exam C items: Q13, Q36, Q68

### Original Exam C Question 13

A REST API returns an HTTP status code of 200.

Which of the following is the most likely reason this code was returned? (Select the best answer.)

- A. The server is acting as a gateway, and a timeout occurred.
- B. The client submitted an incorrect password.
- C. The requested resource has been temporarily moved to another URL.

**Correct Answer: D**

Explanation

Most likely, the requested action was successfully performed on the server if a Representational State Transfer (REST) Application Programming Interface (API) returns a Hypertext Transfer Protocol (HTTP) status code of 200. HTTP status codes are used to provide information to a client when the client requests information from an HTTP server. In addition, HTTP status codes are returned to API clients that communicate with an API by using HTTP, such as a REST API. Both a REST API and an HTTP server that is sending webpages to browsers return an HTTP status code of 200 each time the server successfully performs the client's request. From a web server, this code is typically followed by headers and webpage content. From a REST API, this code is typically followed by headers and output in either JavaScript Object Notation (JSON)-encoded or Extensible Markup Language (XML)-encoded format. An HTTP status code of 302, not 200, indicates that the requested resource has been temporarily moved to another Uniform Resource Locator (URL). The HTTP status code of 302 is typically sent when client submits that is temporarily valid and the a a request by using a given URL no longer server is configured to redirect the client to a temporarily valid URL. This code indicates that the client should continue using the original URL to make these requests, even though the URL is temporarily invalid. An HTTP status code of 301, on the other hand, indicates that a URL has moved permanently and that the client should no longer use the given URL to make requests. An HTTP status code of 401, not 200, indicates that the client submitted an incorrect password. An HTTP server returns a 401 status code if the HTTP server requires authentication to respond to the client's request, but the authentication attempt fails. An HTTP status code of 504, not 200, indicates that the HTTP server is acting as a gateway, and a timeout occurred. Typically, this status code indicates that the HTTP server the client is directly accessing attempted to access a second HTTP server in order to complete the request but was unable to do so. An HTTP status code of 504 can be caused by any number of network errors, including Domain Name System (DNS) problems, congestion, or any issue that might cause a secondary HTTP server to become slow to respond.

### Original Exam C Question 36

Which of the following statements is true about the HTTP PATCH method? (Select the best answer.)

- A. It can be used to implement a webhook.
- B. It can be used to remove an object. O be used to retrieve data from an object.
- D. It can be used to apply partial modifications to an object.

**Correct Answer: D**

Explanation

The Hypertext Transfer Protocol (HTTP) PATCH method can be used to apply partial modifications to an object. The following HTTP methods are most commonly used in Application Programming Interfaces (APIs): © GET - retrieves data from an object © PUT · adds information to an object POST - creates an object e DELETE · removes an object e PATCH - applies partial modifications to an object Although the HTTP POST method is most often used to create a new object or resource, it can also be used webhook. known is a user-defined to implement a A webhook, also as a reverse API, HTTP callback. Webhooks are event-driven rather than request-driven: instead of an application continuously requesting data, the application can wait for a particular event to occur before it reacts. Webhook notifications can be used on many Cisco platforms, including Meraki and Webex Teams.

### Original Exam C Question 68

Which of the following CRUD operations corresponds to the HTTP PUT method? (Select the best answer.)

- A. create
- B. read
- D. delete

**Correct Answer: C**

Explanation

The CRUD update operation corresponds to the Hypertext Transfer Protocol (HTTP) PUT method. The acronym CRUD refers to the four actions that are performed by applications: create, read, update, and delete. allows the formation of variables data Read allows Create new or structures. a client to view data. allows client data. Delete allows client data. The table Update a to change a to remove following shows the relationship between HTTP methods and CRUD operations: HTTP Methods Operations GET Read PATCH Update PUT Update POST Create DELETE Delete The CRUD create operation does not correspond to the HTTP PUT method; it corresponds to the HTTP POST method. The CRUD read operation does not correspond to the HTTP PUT method; it corresponds to the HTTP GET method. The CRUD delete operation does not correspond to the HTTP PUT method; it corresponds to the HTTP DELETE method.

## Day 62 - Software-Defined Networking

Included Exam C items: Q27, Q48

### Original Exam C Question 27

Which of the following Cisco management solutions supports Cisco SDA? (Select the best answer.)

- B. Cisco PI
- C. Cisco 10s 15
- D. Cisco Network Assistant

**Correct Answer: A**

#### Explanation

Of the available choices, Cisco Digital Network Architecture (DNA) Center is the Cisco management solution that supports Cisco Software-Defined Access (SDA). Cisco SDA is a Cisco-developed means of building local area networks (LANs) by using policies and automation rather than by traditional methods of campus management. Cisco DNA Center, which has been rebranded as Cisco Catalyst Center, abstracts and simplifies the complexity of network configuration by implementing a centralized controller and graphical user interface (GUI) to automate the configuration of network devices. Cisco DNA Center also supports many of the same traditional campus device management features that are supported by other Cisco management solutions. Administrators typically interact with Cisco DNA by using a browser-based GUI. Cisco DNA Center uses the Representational State Transfer (REST) Application Programming Interface (API) to natively communicate with Cisco devices. To communicate with third-party devices, Cisco DNA Center relies on software development kits (SDKs). Cisco IOS 15 is not built specifically to support the Cisco SDA. Cisco IOS is a network device operating system (OS) that is used to directly configure, manage, and troubleshoot a single device. Administrators typically interact with Cisco IOS by using a command-line interface (CLI). Access to the CLI can be gained by connecting to a device's console port, by connecting to a Telnet session, or by connecting to a Secure Shell (SSH) session, depending on how the device is configured. Cisco Network Assistant is not built specifically to support the Cisco SDA. Cisco Network Assistant is a free Java-based desktop application that enables a LAN administrator to perform network operations, diagnose problems, and interact with network devices by using a GUI. A typical Cisco Network Assistant installation supports the management of up to 80 devices. Cisco Network Assistant predates Cisco SDA and is therefore not specifically built to support Cisco SDA. Cisco Prime Infrastructure (PI) is not built specifically to support Cisco SDA. Cisco PI is a traditional enterprise Cisco management platform that relies on a browser-based GUI to enable administrators to perform operations on the network, diagnose problems with the network, and interact with devices on the network.

### Original Exam C Question 48

In a controller-based network, the functions of which of the following protocols are most likely to be moved to a centralized controller? (Select 2 choices.)

- A. SNMP
- B. Syslog
- C. sSSH
- D. EIGRP

**Correct Answer: D, E**

#### Explanation

In a controller-based network, the functions of Enhanced Interior Gateway Routing Protocol (EIGRP) and Open Shortest Path First (OSPF) are most likely to be moved to a centralized controller. Routing protocols like EIGRP and OSPF operate in the control plane of a traditional distributed network. These protocols make routing decisions for packets that require routing among Layer 3 devices. In a controller-based network, such as a Software-Defined Networking (SDN) network, the control plane is centralized. Therefore, the decision-making logic for packet-handling policies is either moved to a central controller or monitored by a central controller. In a controller-based network, none of the functions of Secure Shell (SSH), Syslog, or Simple Network Management Protocol (SNMP) are likely to be moved to a centralized controller. All of these protocols operate in the management plane in both a traditional network and a controller-based network. Another network management protocol that

operates in this plane includes Telnet. All of these protocols enable an administrator to connect to and manage a network device. Layer 2 switches, Layer 3 switches, and end devices typically operate in the data plane. In a controller-based network, the controller communicates with the data plane by using a southbound Application Programming Interface (API), such as NETCONF, OpenFlow, OpFlex, or OnePK. Network tasks that are typically performed in the data plane include the encapsulation and decapsulation of packets, the adding or removing of trunk headers, the matching of Media Access Control (MAC) addresses to a MAC address table, the matching of Internet Protocol (IP) addresses to paths in a routing table, the encryption of data, Network Address Translation (NAT), and filtering by using either access control lists (ACLs) or port security.

## Day 63 - Ansible, Puppet, Chef & Terraform

Included Exam C items: Q38

### Original Exam C Question 38

Which of the following does Terraform use as a modeling language? (Select the best answer.)

- A. YAML
- C. SSH
- D. APIs

**Correct Answer: B**

#### Explanation

Terraform uses HashiCorp Configuration Language (HCL) as a modeling language. Terraform is an infrastructure as code (IaC) configuration management tool that can be used to manage cloud-based and local devices. A Terraform configuration file is used to configure the devices. A Terraform resource file specifies the devices, networks, or data stores that you want to manage. Terraform modules are used to organize these configuration and resource files. Terraform does not use Application Programming Interfaces (APIs) as a modeling language. Instead, Terraform uses APIs to communicate with client devices and services. Terraform uses a push model, which means that a Terraform server will push configurations to client devices. Terraform does not use YAML Ain't Markup Language (YAML) as a modeling language. YAML is used by Ansible as a modeling language. Ansible is a configuration management tool that operates on macOS, on Linux, or within a Linux virtual machine (VM) on Windows. An Ansible playbook, which is written in YAML, specifies the actions that will be performed on a client device. Terraform does not use Secure Shell (SSH) as a modeling language. SSH is used by Ansible to securely communicate with client devices. By default, Ansible pushes configurations to clients. However, you can enable clients to pull configurations from Ansible by installing a program called Ansible-pull.